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Mousiness in wine:

**An investigation into the effects of repeated exposures and
systematic training on wine professionals' sensitivity to
mousiness**

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1. Summary

Mousy taint is perceived to be increasing. Variation in wine professionals' (WPs) detection abilities and a lack of consensus regarding the taint represent challenges to sensory quality control. Several studies have found an induction of sensitivity in subjects after repeated exposure to a substance. Thus far, no study has explored whether this also applies to mousiness in wine.

The aim of this research paper was to test whether repeated exposure to one of the mousy compounds, 2-acetyltetrahydropyridine (ATHP), could induce sensitivity in WPs. Fifty volunteer WPs were randomly divided into a training group that conducted systematic training on ATHP, as well as a control group not subject to training. Measurement of detection thresholds were used to monitor changes in sensitivity. The WPs also described two concentrations of ATHP in wine and these descriptions were compared to terms used in the literature.

The results show that WPs' detection thresholds of ATHP can be reduced and exposure to a limited number of clear benchmarks is sufficient to achieve this. Also, the number of subjects truly anosmic to ATHP seems to be much lower than what is commonly referred to in the literature, while the inducible subjects represent a more important category. WPs' descriptors of ATHP also indicated a relationship between retro-nasal odour quality and stimulus intensity, and that the degree to which a mousy wine is unpalatable depends on the concentration.

2. Introduction

Mousiness is a wine fault that has re-emerged in recent years (Tempère, et al., 2019, p. 1) and has even been referred to as “the wine fault of our times” (Goode, 2020). It was described by Thudicum (1894), referenced in Grbin & Henschke (2000, p. 255), as “closely resembling to the smell of a residence of mice,” representing one of the earliest known reports of the fault.

Under normal wine tasting conditions, retro-nasal evaluation is most commonly necessary for detection as the mousy compounds in general are not sufficiently volatile at the low pH of wine, meaning they are first released when the wine comes into contact with an individual’s higher saliva pH (Tucknott, 1977, cited in Costello, 1998, p. 23). It is generally perceived after the wine has been expectorated or swallowed (Snowdon, et al., 2006, p. 6466).

Mousiness is referred to as both unpleasant, persistent and even unpalatable (Grbin, et al., 1996, p. 57), potentially causing significant financial loss to a wine producer. There is, however, a lack of consensus, including among wine professionals (WPs), regarding both the detection and the characterisation of mousy off-flavour in wine (Tempère, et al., 2019, p. 2). Thus, discussions on the topic can be challenging, as a wine may seem faultless to one person yet unpalatable to another.

There have been no recent studies on the actual incidence of mousiness in wine on a larger scale. In contrast, for detection rates of mousiness, one of the most extensive studies to date was undertaken by the Australian Wine Research Institute (AWRI, 2011). The AWRI found that as many as 25% of WPs could not detect the taint,

which makes this wine fault particularly interesting. However, whether this was due to actual specific anosmia or the assessors' lack of familiarity with the taint is unclear.

Several studies have reported an induced sensitivity to odorous compounds following repeated exposure (Tempère, et al., 2012); (Cain & Schmidt, 2002); (Wysocki, et al., 1989). Specific anosmia has for instance been related to the flavour compound diacetyl (Lawless, et al., 1994). However, Tempère et al. (2012) reported a decrease in detection thresholds for both diacetyl and linalool in WPs following a period of training.

Variation in the ability to detect mousiness is partially related to differences in the composition of a person's saliva and the pH of the tongue's surface (Snowdon, et al., 2006, p. 6466). However, if repeated exposure to 2-acetyltetrahydropyridine (ATHP), one of the compounds that produce the taint, could influence WPs' detection ability, these may be effective in reducing both the lack of consensus among WPs and increase their sensory quality control ability to detect mousiness under normal wine tasting conditions.

This research paper (RP) will therefore investigate the following:

- 1) What are wine professionals' detection levels of ATHP pre- and post-exposure?
- 2) Can wine professionals' sensitivity to mousiness be changed by repeated exposures and systematic training?

- 3) What are wine professionals' descriptors of one of the mousy compounds, ATHP, in wine and how do these relate to established terms by previous literature?

Given the apparent increase in the incidence of mousiness and the high degree of specific anosmia reported by the AWRI (2011), exploring whether repeated exposures to mousiness can improve WPs' detection ability may be of significant value to the wine trade.

3. Background and context

3.1 Mousiness in wine

3.1.1 Occurrence of mousiness in wine

Historically, mousiness has been described as both “sporadic” (Snowdon, et al., 2006, p. 6465) and “infrequent” (Grbin, et al., 1996, p. 57), though these authors acknowledge the strong detrimental effect on wine quality when mousiness does occur.

However, more recently, both wine experts and researchers share an impression that the presence of the taint is increasing (Tempère, et al., 2019, p. 1); (Kiyomichi, et al., 2023, p. 1); (Snowdon, et al., 2006, p. 6465); (A. Feiring, 2017, personal communication, 22 October). Indeed, several articles on the topic have recently been published in the international wine press (Woolf, 2017); (Goode, 2020); (Feiring, 2022). Further, the AWRI (AWRI, u.d.) reported that an increasing number of wine samples suspected of mousiness had been submitted to its technical services group in the 1990s and 2010s.

The rise of the natural wine movement over the past two decades may be of relevance to explaining the apparent increase in mousy wines. Massini & Vuchot (2015) associated mousiness in wines with a significant reduction in the use of SO₂ (supported by recent research by Kiyomichi, et al., 2023), increasing pH levels, and the use of native microbiota in fermentations. Thus, the fault is particularly relevant to natural wines made with minimal intervention. Another explanation for the increased

reported incidence of the wine fault may be increasing awareness and familiarity with the taint among WPs.

3.1.2 Formation of mousiness

Mousiness is usually of microbial origin (AWRI, u.d.) and is mainly attributable to Dekkera/Brettanomyces yeast (Grbin & Henschke, 2000) and certain strains of lactic acid bacteria (Costello, et al., 2001). There is no method for removing mousy off-flavour once it has occurred, but an effective form of protection against the taint seems to be SO₂ (AWRI, u.d.).

Three *N*-heterocyclic molecules have been associated to the mousy off-flavour in wine: 2-acetyltetrahydropyridine (ATHP) (Strauss & Heresztyn, 1984), 2-ethyltetrahydropyridine (ETHP) (Grbin, et al., 1996, p. 59) and 2-acetyl-1-pyrroline (APY) (Herderich, et al., 1995). Of the three, EHP was previously considered the least important compound. In fact, its sensory threshold measured in water is higher than both ATHP and APY (Tempère, et al., 2019, p. 2). Nevertheless, it is difficult to compare the detection threshold values per se, especially when the compounds are diluted in a complex matrix such as wine (Kiyomichi, et al., 2023, p. 2).

Research by Hayasaka (2019) and Kiyomichi, et al. (2023) recently provided reliable and rapid methods for the quantification of the *N*-heterocyclic compounds associated with mousiness, which can be useful for the routine screening of wines. In fact, the latter showed that, based on a sample of 68 wines produced with limited or no SO₂, EHP was found in almost all and ATHP was found in nearly all wines suspected of

mousiness. In contrast, APY was only found in 11 wines and therefore represented only a few cases compared to AHP and EHP. Also, in Strauss & Heresztyn's research (1984), AHP was only present in mousy and not sound wines.

3.1.3 Sensory character and the detection ability of individuals

Various research referenced in Tempère, et al. (2019, p. 2) described a mousy character in wine as similar to rodent urine (dirty mouse cage) and grilled foods, such as popcorn, rice, crackers and bread crusts (Tucknott, 1974); (Buttery, et al., 1983); (Strauss & Heresztyn, 1984); (Herderich, et al., 1995). Other descriptions include dried sausage skin, vomit and dirty mops (Tempère, et al., 2019, p. 2). The wide range of descriptions may be due to the potential and unexplored matrix effect between the compounds involved, and their relative levels of intensity and variation in individual detection abilities (Tempère, et al., 2019, p. 2).

Ever since the early 1900s, variations in individuals' ability to detect mousiness have been recognized (Müller-Thurgau & Osterwalder, 1913). Using AHP, Grbin, et al. (1996), examined individuals' detection thresholds and concluded that their range of detection was large.

APY, EHP and AHP are generally not sufficiently volatile at the pH range of wine to be detected ortho-nasally (Bartowsky & Henschke, 1995). However, when the wine (2.8–3.8 pH) comes into contact with the higher pH range of a person's saliva (usually near 7), the off-flavour can be detected retro-nasally. Nevertheless, there are

wide inter-individual variations in oral pH of between 5.76 and 7.96 (Larsen, et al., 1999), which may partially explain the differences in individuals' ability to detect mousiness (Snowdon, et al., 2006, p. 6466).

3.1.4 Evaluation techniques

Ortho-nasal evaluation techniques have been developed to detect mousy off-flavour in wine without actually tasting it, such as the “palm and sniff” method, where one drop of wine is placed on the back of the hand and the skin is then sniffed (Grbin, et al., 1996, p. 58); the alkaline strip method, using alkaline strips dipped in culture media or wine (Grbin & Henschke, 2000); oxidation of the wine due to the instability of APY and ATP in its presence (Weerawatanakorn, et al., 2015); and pH adjustment (Tempère, et al., 2019). Tempère, et al. (2019) found that by adjusting the pH to around 5, the consensus among WPs increased significantly, as well as their ability to discriminate. These methods, however, are not representative of normal wine tasting conditions.

3.1.5 Previous research

Important wine literature, such as the Oxford Companion to Wine, has often stated that as many as 30% of WPs cannot detect mousiness. The source of this specific figure is not cited. However, this percentage is also widely reported elsewhere (Feiring, 2022); (Goode, 2018, p. 180); (Woolf, 2017).

The 30% anosmia figure likely originated from the wine taint and fault tasting clinics organized by the AWRI during their Advanced Wine Assessment Courses (A.

Coulter¹ 2022, personal communication, 18 February) and is also referred to in “Ask the AWRI column” of Grapegrower & Winemaker (2015). In accumulated data from 2006 to 2011, only 75% of WPs could detect mousiness (AWRI, 2011). As the anosmia data were collected over time, the 30% figure often quoted in various sources of wine literature was based on 148 study participants and not the final number of 238 participants who accounted for a lower average anosmia of 25% (A. Coulter 2022, personal communication, 18 February).

These results, however, were based on one single and unquantifiable concentration of ATHP presented to the study participants and did not take into account different levels of sensitivity. Thus, rigorous sensory investigations to collect the anosmia data were not conducted (A. Coulter 2022, personal communication, 18 February). Because of its prominence, the AWRI’s study is regarded as important research.

3.2 Sensory evaluation and measurement of detection thresholds

In this RP, sensory evaluation is used to determine the level at which the stimulus of mousy compounds is detected and if it can be changed through repeated exposures. The 50 WPs of the RP are referred to as “assessors” and their detection levels as “thresholds”. These are taken as a description of the assessors’ performance.

Three-alternative forced-choice (3-AFC) is a common procedure for measuring odour, flavour and taste detection thresholds (ISO 13301, 2018), and was used in the RP. During a 3-AFC test, the participants of a panel are presented with three

¹ Senior Oenologist, AWRI

samples, one of which contains the substance to be detected (test sample), while the other two samples (references) do not contain the substance in question. The participants are asked to choose the test sample, and if uncertain they are required to guess (ISO 13301, 2018).²

An ascending series is mostly used in taste and smell to avoid adaptation and carry-over effects from a strong concentration affecting a successively weaker one (Lawless & Heymann, 2010, p. 21). Lawless & Heymann (2010, p. 19) refer to sensory adaptation as “a commonly observed decrease in responsiveness to conditions of more or less constant stimulation.” The potential effects of adaptation must therefore be anticipated in the experimental design of a sensory study (O'Mahony, 1986) and are highly relevant when taking into account the persistent character of the mousy compounds and the need for retro-nasal detection (S. Tempère³ 2023, personal communication, 13 October).

3.2.1 Criteria for a correct answer

Because 3-AFC testing requires the participants of a panel to guess when they are uncertain, the literature has taken different approaches to the kind of criteria that should apply to a correct answer when establishing individual detection thresholds. Lawless and Heymann (2010, p. 131) refer to a participant's first correct answer when all higher concentrations are also correct. At the same time, they (p. 137)

² In a 3-AFC test where only one of three available choices is correct, the probability of selecting the correct sample by chance is equal to 1/3 (ISO 13301, 2018).

³ Assistant Professor at the Institute of Vine and Wine, University of Bordeaux

acknowledge the probability of the potential effects of adaptation in the threshold tastings where “the sensory impression will sometimes “fade in and out” at levels near threshold.” Furthermore, a strong stimulus late in the ascending series may be perceived as weaker than it actually is, due to the repeated exposure to a given substance.

Thus, several studies such as Tempère, et al. (2011, 2012, 2019) and Wise, et al. (2008) have applied as criterion for a correct answer the lowest concentration detected of three correct answers in a row. The latter study also asked the study participants to confirm their certainty regarding the last correct answer of the three (based on the former version ASTM E679-04). With such an approach, however, there is a risk of a) interference between detection and recognition thresholds (S. Tempère 2023, personal communication, 13 October) and b) influencing the results with a subjective component as the group of assessors will likely comprise both conservative and more liberal individuals with regards to their ability to be certain (Macmillan & Creelman, 1991, referenced in Wise, et al., 2008, p. 17).

Another method for minimising the effects of adaptation is to use a stopping rule⁴. However, this approach could also lead to an incorrect result (W. Pearson⁵ 2024, personal communication, 1 February) and miss important data for a subsequent probability of detection analysis. Considering the above, the most appropriate

⁴ A stopping rule means that individuals may stop tasting after giving a predefined number of correct answers (Lawless & Heymann, 2010, p. 133).

⁵ Senior Research Scientist, AWRI

criterion for a correct answer will likely depend on the substances that are to be detected.

4. Methodology

4.1 Pilot Study

In 2022, an in-depth pilot study was conducted involving 30 WPs from the Norwegian Wine Monopoly (Vinmonopolet). All the pilot study participants performed two ascending 3-AFC tests based on eight different combinations of two of the mousy compounds: ATHP and APY, with a period of 13 days of training in between the two tasting sessions. The concentration ranges of APY and ATHP (Table 2) were based on the concentrations found in spoiled wines in the previous literature (Grbin, et al., 1996, p. 59), as well as preliminary experiments by Laboratorios Excell Ibérica S.L. and the researcher.

The results indicated that, on average, the pilot study participants performed better in the second 3-AFC test. However, some of the results also raised questions about whether training sometimes resulted in certain individuals adapting, leading to a reduced level of sensitivity after training. Also, some of the pilot study participants seemed to be affected by adaptation and/or fatigue during the two tasting sessions and therefore questioned a) whether the minimum 45 seconds that elapsed between each wine tasted in the 3-AFC tests were sufficient and b) the appropriateness of using the first correct concentration identified where all other concentrations are also correct (Lawless & Heymann, 2010, p. 131) as the criteria for a correct answer for the best estimate of individual threshold (BET).

12 pilot study participants were able to detect all concentrations in the first tasting session, showing that the range of concentrations was probably too high for a 3-AFC test. As the interactions between ATHP and APY are unknown, the effect of each of the compounds was also difficult to interpret. Furthermore, due to the lack of a fixed geometrical dilution factor between the concentration steps, detection thresholds could not be estimated in accordance with ISO 13301 (2018). Finally, it could be questioned whether the average improvement of the pilot study participants was due to repeated exposure from repetition of the 3-AFC test, or to the training.

4.2 Materials and methods

4.2.1 Experimental design

Based on the experience from the pilot study, the new RP study had a higher number of assessors (50) to increase the statistical robustness of the research. For the threshold measurements, all assessors were required to complete two identical sensitivity tastings using ascending 3-AFC tests referred to as T1 (first sensitivity tasting) and T2 (second sensitivity tasting). Only ATHP was used, with a consistent dilution factor between each concentration step. A minimum break of two minutes between each wine tasted was required for the 3-AFC tests.

All assessors were randomized to a control group (C) and a training group (T) to distinguish between the effect of repeated exposure through 3-AFC compared to systematic training. In between T1 and T2, T trained at home for 14 days. Estimated detection thresholds for ATHP were used to measure changes in sensitivity between T1 and T2 at individual and group level.

The criterion used for assessing BET for T1 and T2 was set to be the lowest concentration detected of three correct answers in a row in line with the research referenced in 3.2.1. This criterion would therefore consider the potential effects of adaptation that are especially relevant to a persistent taint such as mousiness (also indicated by the pilot study), while not potentially losing important data by applying a stopping rule. If an assessor did not reach the criterion of three correct answers in a row but answered correctly at the end of the series (e.g. the last two concentration steps), a correct answer was considered the first of those correct answers following research of Wysocki, et al. (1989).

The probability of finding three correct answers in a row at random is only 3.7% (S. Tempère 2023, personal communication, 13 October). Despite the low likelihood of making three correct guesses in a row, an additional and stricter criterion compared to the referenced research was applied: three correct answers in a row were only considered correct when the assessor had a maximum of one incorrect answer later in the series. As one incorrect answer after three correct answers may occur as a consequence of adaptation explained in 3.2.1, several incorrect answers would, in contrast, suggest that the assessor had not accurately identified three in a row, but had instead guessed correctly.

4.2.2 Location and schedule

All tastings took place at Vinmonopolet's headquarters in Dronning Eufemias gt. 13, Oslo, Norway. The test room was designed in accordance with the *general guidance for the design of test rooms* (ISO 8589, 2007) and the standard for *Testing and*

calibration laboratories (ISO/IEC 17025, 2017) with seven separate tasting booths and a room temperature of 19–21 °C. All booths included an integrated spittoon and PC.

All tastings were performed between 09:30 and 17:00. Due to the high number of assessors and the capacity of the test room, both T1 and T2 were organized over a period of two days each.

4.2.3 Assessors and panel size

The cohort of this study comprised 50 WPs from different sectors of the Norwegian wine trade. Their participation was based on their own desire to learn. In the research by Parr, et al. (2002), the expert group comprised winemakers, wine science researchers, wine professionals (Masters of Wine, wine judges, wine writers, wine retailers), graduate students of viticulture and oenology and people who had been involved in wine for more than ten years.

Due to the limited amount of wine produced in Norway, the most suitable assessors were employees of Vinmonopolet (retail), wine importers, wine educators and wine writers. All the assessors had completed a minimum WSET 3/Sommelier degree, were currently working full time in the wine trade with regular wine tasting as part of their work and had not been part of the pilot study. Twenty-seven female and 23 male assessors participated in the study with ages ranging from 18–60+ (Table 1).

Lawless & Heymann (2010, p. 130) consider a panel of 25 assessors to be sufficient for threshold tastings, while Leske, et. al. (2013, p. 23) suggest a minimum of 15–20. For an equivalent study, Tempère, et al. (2012) had a total of 32 WPs divided into a training and a control group comprising 16 WPs each. Since the pilot study did not yield the desired robustness due to the elimination of some hyper-sensitive assessors, a notably higher number of assessors than the pilot study, as well as similar studies and the recommendations described above, were targeted for the RP.

Age range	Male	Female	Total
18–29	2	0	2
30–39	7	7	14
40–49	6	15	21
50–59	7	4	11
60+	1	1	2
Total	23	27	50

Table 1: WP demographics

4.2.4 Pre-screening of assessors

In advance of T1, the assessors were asked whether they had previously detected mousiness in wine. They were also subject to a pre-screening of their detection ability. This took place a minimum of one hour before T1. The assessors were presented with three glasses of wine: one reference wine (W1) with no added substance and a set of two wines (W2 and W3) in which one was spiked with the highest concentration of ATHP (C8, Table 2) and the other was not. In their own time, they were asked to describe W2 and W3 and whether/how each of them was different from the reference wine.

The purpose of the pre-screening was as follows:

1. To ensure that all assessors had the exact same benchmark of the compound to be detected during the subsequent 3-AFC tests.
2. To give an indication of potentially anosmic assessors who could then be randomized equally into C and T.

4.2.5 Sample preparation and glassware

Wine and samples were prepared in a separate room away from the assessors. Two hours prior to the tastings, bottles of the base wine were spiked with the different concentrations of mousy compounds (Table 2) supplied by Laboratorios Excell Ibérica S.L. in small vials. Each vial contained mousy compounds precisely measured for a 75cL bottle. 50mL of base wine was measured for each glass, using a graduated measuring device on top of each bottle. The pilot study wine was slightly oaky⁶ with a light popcorn note. As popcorn can describe mousiness, this could have been a confounding factor in the results of the pilot study. Thus, a neutral unoaked Chardonnay from the Languedoc⁷ was chosen as being more appropriate for the RP. All glasses were *Spiegelau Special Glasses Expert Tasting*, which are also used for Vinmonopolet's blind tastings⁸.

⁶ *Lindeman's Bin 65 Chardonnay* was regularly used in the AWRI's previous research (2011) (A. Coulter 2022, personal communication, 18 February) and was therefore used in the pilot study.

⁷ Vintage 2022, no malolactic fermentation, 12.3% alcohol, pH 3.35.

⁸ The glasses are similar to ISO 3591 wine tasting glasses but have a slightly wider diameter.

4.3 Stimuli

4.3.1 Concentration ranges

Due to ATHP's relevance in the occurrence of mousiness and its role in previous research, this compound was selected for the concentration range.

The concentrations of ATHP were determined based on the concentrations measured in spoiled wine by Grbin, et al. (1996, p. 59) from 4.8µg/L to 106µg/L, Costello (1998, p. 95) from 0.7µg/L to 106µg/L and Hayasaka (2019, p. 114) from 0.52µg/L to 26.46µg/L, as well as the results from the pilot study (Table 2).

The final levels of the concentrations selected were tested by using a 3-AFC test, performed under the same conditions and circumstances as in the main study of the RP, on a selection of 13 participants from the previous pilot study (described in 4.1), including the least and most sensitive participants. None of them could detect the lowest concentration of 0.2µg/L, and only one of the participants could not detect the highest level of 122.07µg/L. Thus, the selected concentrations seemed appropriate to cover a variety of thresholds.

A dilution factor of 2.5⁹ was applied between each of the eight concentration steps for the successive BET calculations. Due to the requirement for retro-nasal evaluation and taking into account the results from the pilot study, eight steps with no

⁹ A dilution factor of two or three is common in taste and smell where the concentrations represent equal steps on a log scale (Lawless & Heymann, 2010, p. 130).

replications were considered the maximum number of steps due to the risk of palate fatigue and adaptation. However, these were also considered sufficient for a threshold tasting (Lawless & Heymann, 2010, p. 130).

Concentration (C)	RP	Pilot study	
	ATHP in µg/L	APY in µg/L	ATHP in µg/L
C1	0.20	0.05	1
C2	0.50	0.1	2
C3	1.25	0.3	3
C4	3.13	0.5	4
C5	7.81	1	6
C6	19.53	3	25
C7	48.83	5	50
C8	122.07	8	100

Table 2: Concentrations used in the 3-AFC tests of T1 and T2 of the RP (including the systematic training procedure for T and pre-screening of all assessors) and the pilot study¹⁰.

4.4 Procedure

4.4.1 Sensitivity measurement tastings (T1 and T2)

T1 and T2 were completely identical.

The assessors were instructed beforehand to avoid consuming anything but water one hour prior to the tastings. Each assessor was provided with a set of 24 glasses of

¹⁰ In the pilot study detection thresholds could not be estimated in accordance with ISO 13301 (2018) due to the lack of a fixed dilution factor between the concentration steps. The concentrations of the RP were therefore different and also covered a wider range of detection thresholds. References to C1–C8 throughout the RP refer to the concentrations of the RP and not the pilot study.

wine at a temperature of 19–21°C, presented simultaneously. The samples were all coded with three digits and distributed at random between the assessors. The 24 glasses of wine were divided into eight trials presented to the assessors as one line for each trial, starting with the line containing the lowest concentration (C1) of mousy compounds (closest to the assessors) and finishing with the strongest concentration (C8), which was present in trial eight (furthest away from the assessors). All samples were tasted from left to right. The assessors submitted their answers online in EyeQuestion (EyeQuestion v.4.11.70, 2018). They were instructed to taste each sample only once to avoid fatigue. They were not informed of the results during the tasting, nor that the trials were increasing in intensity.

The assessors were asked to expectorate the wine after four to eight seconds and wait a minimum two minutes before tasting a new wine to avoid residual mousiness affecting a subsequent clean sample. The results from the pilot study revealed that the lowest concentrations of mousiness could first be perceived as long as 16–20 seconds after the wine had been expectorated. However, some of the assessors perceived the strongest concentrations while the wine was still in their mouths and had a lengthy persistency.

The suggested minimum break of two minutes between each wine proved to be sufficient to avoid the effects of adaptation based on the pre-test with 13 of the participants from the pilot study described in 4.3.1. The assessors were also encouraged to wait even longer if necessary to allow sensory recovery between wine samples containing high concentrations (ISO 13301, 2018). The assessors were

permitted to rinse/neutralize their palates with neutral table water and biscuits in between the wines tasted. 30 minutes before the tastings, the assessors were informed both verbally and in writing about the tasting procedure, as explained above.

4.4.2 Training

The assessors of T were asked to follow a systematic training programme individually at home for two weeks in between T1 and T2. Previous sensory training research on other compounds has included training periods with different levels of intensity lasting from five to 60 days (Tempère, et al., 2012); (Wysocki, et al., 1989); (Cain & Schmidt, 2002); (Dalton, et al., 2002). The 14 days proposed for this RP were within this range.

All assessors completed T1 and T2. However, any assessors who detected all concentrations of AHP in T1 were excluded from assignment to C or T, in accordance with research by Tempère, et al. (2012). Nevertheless, ensuring that these hyper-sensitive assessors still completed T2 allowed a qualitative exploration of potential variations in the thresholds of such WPs. Assessors who could not detect C8 during T1, on the other hand, were randomized equally into C and T.

After completing T1, the assessors assigned to T were provided with four 5.7cL Vivaplex glass bottles of the base wine to taste over the next 14 days. Two of the wines were spiked with the strongest concentration of mousiness (C8) and the

remaining two with C4 (Table 2). Small bottles were selected for practical reasons so that the assessors could carry the bottles as hand luggage on planes, etc. The training kit also included a vial with residues of C8, dissolved in ethanol, to nose daily during the training period which, due to the higher pH, could be detected ortho-nasally.

The results from the pilot study revealed that some participants felt less sensitive to mousiness following the training period when they tasted both concentrations every day. Thus, for the RP, the assessors in T were asked to alternate their training with C8 and C4 over the two weeks. To track their commitment to the training, they would access EyeQuestion each day by scanning a QR code attached to the bottles on their phone and indicate on a scale from 1–9 how they had perceived the concentration they had tasted each day (1=no mousiness, 9=strong mousiness). Each assessor trained for an average of 6.8 days for both C8 and C4 with a minimum of five days and a maximum of seven days for each.

4.4.3 Qualitative descriptions of mousy samples

After completing T2 and a break of a minimum of 20 minutes, the assessors were briefly asked to describe two glasses of wine containing C4 and C8 respectively, without knowing either the compound or the concentrations used and were given a reference sample with no added substance for comparison. As a final step, T assessors were also asked to report how the training had affected their personal acceptance, perceived sensitivity, level of certainty and awareness of mousiness in wine.

4.5 Measurement of detection thresholds

4.5.1 Probability of Detection

Probability of Detection (PD) of the panel as a whole was calculated for T1 and T2 in order to find the absolute detection level as the concentration where PD=0.5, thereby indicating the concentration level at which 50% of the panel had successfully detected ATHP. PD was modelled as described in ISO 13301 (2018) by considering all data from the threshold tastings, adjusting the data for chance, taking into account the guessing model represented by 3-AFC, in which P_c represents the probability of making a correct selection¹¹:

$$PD(\text{concentration}) = 1.5 * P_c(\text{concentration}) - 0.5$$

For comparison of the results, a linear model was made with group (categorical variables with levels C and T), test (categorical variables with levels T1 and T2) and $x = \log_{10}(\text{concentration})$ (continuous variable) as independent variables and PD as the dependent variable, including all interaction terms. A variable selection was performed with stepwise selection using the Akaike information criterion (AIC)¹² using the R-function step with default settings (Venables & Ripley, 2002) to obtain the most parsimonious model. The final model included only main effects, resulting in regression models with different intercepts defined by group (C, T) and test (T1, T2). The adjusted R-square is reported for both the full and reduced model. Analysis of

¹¹ In a 3-AFC test a correct answer may happen by chance, especially at lower concentration levels, since the assessors are required to guess if uncertain as explained in 3.2. ISO 13301 (2018) describes in detail a common mathematical model applied to take into account the probability of guessing correctly (1/3). According to this standard, the detection threshold is defined as the concentration level where 2/3 of the panel makes a correct selection, which further corresponds to the probability of detection=0.5.

¹² AIC is a measure of model fit

Variance (ANOVA) was used to evaluate whether there were significant differences at the panel level between T1 and T2, and whether there were significant differences between the groups (C, T).

4.5.2 BET measurements

Individual BET estimates and group thresholds, taking the geometric mean¹³ of the individual values (ISO 13301, 2018), were appropriate for a more in-depth analysis.¹⁴

In accordance with ISO 13301 (2018), BET was calculated as the geometric mean between the last concentration that was missed and the first correct concentration. For assessors who gave the correct answer at the lowest proposed concentration, the threshold was identified as the geometric mean between this concentration and the hypothetical next lowest calculated concentration. For assessors who failed to correctly recognise the highest concentration, the threshold was defined as the geometric mean between the highest proposed concentration and the subsequent hypothetical calculated concentration.

¹³According to Lawless and Heymann (2010, p. 129) the geometric mean is a reasonable choice when it comes to finding the minimum level of a substance that is detected by 50% of a sample group, as it is often very close to the median of a positively skewed distribution, which is often a consequence of high outliers caused by insensitive individuals in the threshold data.

¹⁴ PD was used for calculating group thresholds at panel level. BET estimates were also useful to evaluate individual performances. Both methods are based on 3-AFC and are suggested in ISO 13301 (2018) and were therefore considered relevant. Both take into account the probability of correct guesses in a 3-AFC test (PD by following the mathematical model explained in ISO 13301 (2018); BET by the criterion for a correct answer explained in 4.2.1).

BET values were also analysed after log transformation (base 10), and the $\log_{10}(\text{BET})$ will be referred to as logBET.

4.6 Statistical analyses

Pairwise tests of BET and logBET, with pairwise observations resulting from T1 and T2 within each group, were analysed using the Wilcoxon signed-rank test.¹⁵

The change between T1 and T2 was measured by the difference in the number of concentration steps, the fold change (FC) computed as $\text{BET}_2/\text{BET}_1$, and the log fold change (LFC), which is the logarithm (base 10) of the FC. The values of C and T were compared using Wilcoxon rank-sum tests.

Non-parametric tests were used as the data did not follow a normal distribution according to the Shapiro-Wilk test.

It is reasonable to expect that the improvement potential will vary depending on an assessor's sensitivity. Thus, the assessors on the panel (independent of C or T) were

¹⁵ The Wilcoxon signed-rank test is known to be an alternative for paired tests when normality does not hold. The test does not assume a specific statistical distribution and rely only on the assumption that the distribution is symmetric around a central value and aims to test if this is significantly different from zero. That is, the test considers if the difference between paired observations (the two BET estimates obtained in T1 and T2) are different or not. The Wilcoxon signed-rank test has more power than the sign test because it considers the magnitude of the differences. Likewise, Wilcoxon rank-sum test was chosen to compare LFC for C and T, as this is a two-sample comparison. Wilcoxon rank-sum test is also known as the Mann-Whitney U test.

categorized as high sensitive or less sensitive based on the results of T1. The criterion set for high-sensitive assessors (H) was the correct identification of C4 and lower, while the remaining assessors were categorized as less sensitive (L). There were 13 (H) and ten (L) in C and 11 (H) and 12 (L) in T.

ANOVA was used to consider the effect of the sensitivity group (L, H) and group (C, T), as well as the interaction between them, with LFC or rank (LFC) as the dependent response variable. Ranking was used as the data did not meet the assumptions of normality. As the sensitivity groups were not perfectly balanced, an F-test type III was used for significance testing.

In a separate analysis, the same approach, including BET (T1) as the dependent response variable, was used to compare those assessors who, in the pre-screening, reported they had experienced mousiness before with those assessors who had not/were uncertain. Comparisons were also made using Wilcoxon rank-sum tests.

In all the analyses, the significance level was set to $\alpha=0.05$. All statistical data analyses were performed with R (R Core Team, 2022) by the Norwegian Food Research Institute (Nofima) on behalf of the researcher. Subsequent interpretation and critical analyses were conducted by the researcher.

5. Results and analysis

The threshold measurements showed variation both across assessors and between T1 and T2. Based on T1, four of the total 50 assessors were classified as hyper-sensitive according to the criterion explained in 4.4.2. Therefore, C and T consisted of 23 assessors each. All raw data are presented graphically in the Appendix.

5.1 Research Question 1

What are wine professionals' detection levels of ATHP pre- and post-exposure?

PD and BET measurements served to answer this question.

5.1.1 Probability of Detection

The model fitted for each group and test is presented in Table 3, where the estimated concentration for $PD=0.5$ is also given. For the panel as a whole there was a significant difference between T1 and T2 ($p=0.03$) indicating that, with more than 95% certainty, repeated exposure can induce sensitivity to ATHP in WPs.

A decrease in detection threshold was observed for both C and T from T1 to T2 (Table 3). The level of sensitivity in T was somewhat lower in T1 compared to C ($p=0.07$) and this difference persisted in T2. No interaction between test (T1, T2) and group (C, T) was reported, meaning that changes from T1 to T2 were not significantly different between C and T.

Figure 1 shows the proportion of assessors in C and T who gave correct answers for each concentration in T1 and T2. With limited exceptions, such as C4 for both groups and C7 for T, the number of correct answers was higher for each concentration in T2 compared to T1.

Group	Test	Model	x	cc	Concentration step
C	T1	$PD = 0.29 + 0.32x$	0.64	4.37	4–5
C	T2	$PD = 0.42 + 0.32x$	0.25	1.77	3–4
T	T1	$PD = 0.19 + 0.32x$	0.96	9.12	5–6
T	T2	$PD = 0.32 + 0.32x$	0.57	3.70	4–5

Table 3: Estimates where $PD = 0.5$, $x = \log_{10}$ concentration, cc = concentration where $PD = 0.5$, concentration step = the relative concentration steps.
C $n=23$; T $n=23$.¹⁶

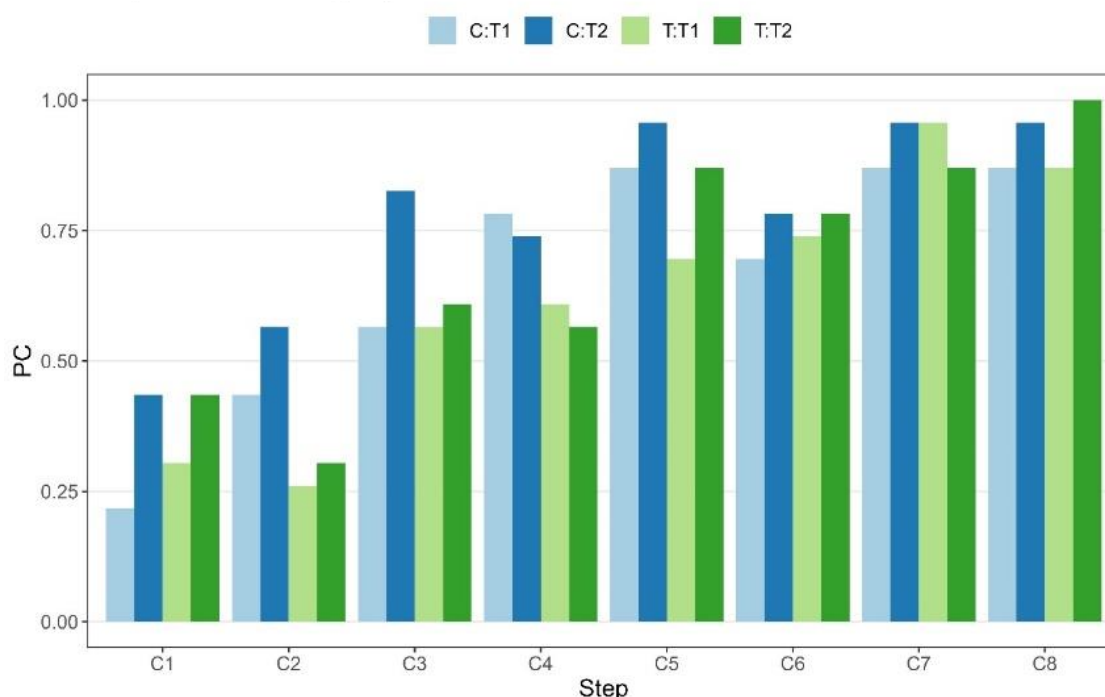


Figure 1: Proportion of assessors who gave a correct answer for each concentration in T1 and T2. C $n=23$; T $n=23$.

¹⁶ Adjusted R-square was 0.80 for the full model and 0.82 after step wise variable selection.

5.1.2 BET measurements

The individual thresholds for AHP covered the whole range of concentration steps and beyond (Figure 2) and the results showed an improved or equal sensitivity for most of the assessors on the panel in T2 compared to T1. Nevertheless, five assessors in T and two assessors in C decreased their performance from T1 to T2. However, three of the four hyper-sensitive assessors, none of whom had been subject to training, also decreased their performance.

Both C and T had lower mean values for BET and logBET in T2 compared to T1, indicating an average increase in sensitivity in both groups (Table 4). On average, the whole panel improved by 0.7 steps, C by 1.4 steps and T by 0.5 steps.

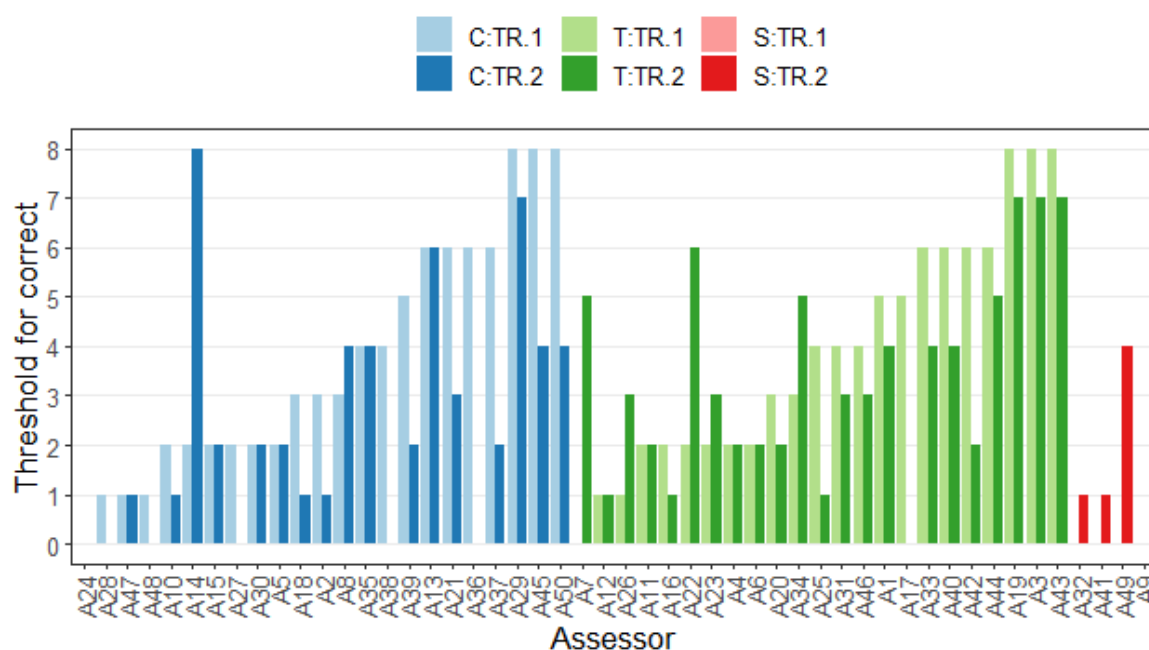


Figure 2: Assessors' individual thresholds in T1 (TR.1) and T2 (TR.2). TR = Last incorrect concentration before first correct of three correct answers in a row. C n=23; T n=23; S n=4 (Hyper-sensitive assessors who were not randomized to C or T).

	Mean whole panel	Mean C	Mean T
BET T1	29.77 (61.85)	32.02 (64.74)	32.68 (64.46)
BET T2	13.24 (33.48)	14.30 (42.41)	14.23 (25.87)
LogBET T1	0.49 (1.00)	0.57 (0.98)	0.66 (0.95)
LogBET T2	0.21 (0.89)	0.04 (0.93)	0.47 (0.82)
Diff	0.72 (2.39)	1.35 (2.42)	0.48 (2.25)
FC	13.23 (37.54)	20.34 (50.83)	8.35 (21.15)
LFC	0.29 (0.95)	0.54 (0.96)	0.19 (0.90)

Table 4: Summary of mean BET, logBET and values for differences including standard deviations. Mean whole panel n=50; Mean C n=23; Mean T n=23. Diff = Average improvement in concentration steps.

5.2 Discussion Research Question 1 results

As Figure 2 clearly shows, there is wide range in individual detection thresholds for ATHP, which is in line with previous research on mousy compounds (Grbin, et al., 1996). The selected concentration range therefore appears to have been appropriate to cover the wide range of thresholds.

5.2.1 Hyper-sensitive assessors

Following T1, four assessors were classified as “hyper-sensitive”. To guard against the effect of correct guesses at the lowest concentration, these assessors performed a retest of the first trial, where they were all able to successfully detect C1 again.

Interestingly, during T2, only one of these assessors could detect C1, and one of them even failed to detect C3 and C4. After the experiment, the latter assessor (A49) explained that she had felt confident about all concentrations during T1, while they

seemed more difficult during T2. This is line with research by Stevens, et al. (1988), where the individual thresholds of assessors in the study often varied from one test to another, despite stable average thresholds based on several measurements.

5.2.2 Least sensitive assessors

Two of the 50 assessors failed to detect C8 during the pre-screening. The results from T1 revealed that the same two assessors as well as four more assessors did not detect C8, giving a total of six assessors characterized as the least sensitive in the first phase of the study.

In T2, C8 was detected by all six assessors referred to above. Considering the possibility of correct guesses, after completing the 3-AFC test, all six were asked to do a retest of the last trial where C8 was again detected successfully by all of them. When asked about their level of certainty before revealing the correct result, they all confirmed 100%.

In contrast, one assessor in C (A14) detected C8 in both the pre-screening and from C3–C8 during T1, while showing unstable results during T2, also failing to detect C8. However, when subject to a retest of the highest dilution step, this assessor also detected C8 successfully and confirmed 100% certainty. Thus, the assessor was able to detect C8 in three out of four sessions. The variation in this assessor's performance in T2 was therefore more likely a consequence of palate fatigue or adaptation, variation of individual thresholds from one test to another (Stevens, et al.,

1988), or even self-induced psychological pressure of performing a second test in which the repeatability could lead to a decrease in performance due to overthinking and limited experience of an in-depth 3-AFC test (W. Pearson 2024, personal communication, 3 April). Due to the instability of the responses of this assessor during T2, the retest of C8 was not considered a correct answer in the BET measurements, but rather serves as an indication that anosmia with regards to ATHP could not be confirmed in this panel of 50 WPs.

5.2.3 Specific anosmia

A previous study (2011) by the AWRI found that 25% of WPs could not detect mousiness. This study was based on a significantly higher number (238) of participants than this RP (50). However, as the concentration of ATHP in the AWRI's study could not be determined, the results did not reflect the different ranges of sensitivity in the participants as some of them simply might have been less sensitive to the concentration that was presented to them.

Based on the above and what concluded in 5.2.2, the probability of having no anosmic assessors in the panel was calculated by a binomial probability distribution with the parameters $p=0.25$ and $N=50$. The probability that none of the 50 assessors were anosmic was highly significant ($p<0.001$).

If a softer definition of anosmia was taken, it could be said that the seven least sensitive assessors who had problems detecting C8 during either T1 or T2

(described in 5.2.2) could be categorized as anosmic. However, even with this looser definition the estimated proportion of anosmic assessors was only 14%, which is significantly lower ($p=0.04$) than the 25% from the AWRI study.

5.3 Research Question 2

Can wine professionals' sensitivity to mousiness be changed by repeated exposures and systematic training?

Research Question 2 further examined the differences between C and T.

5.3.1 Improvement between T1 and T2

21 and 18 assessors in C and T respectively had a similar or better performance in T2 compared to T1, and in both groups 14 assessors achieved a better result (61%). Lower BET was registered for both C and T in T2 compared to T1, and the level of improvement is shown in Table 4.

Considering pairwise tests with BET and logBET as response variables (Table 5), there were highly significant lower values in T2 compared to T1 (Table 5) for C ($p\leq 0.01$). For T the values were significantly lower for BET ($p<0.05$) while logBET showed a weaker significance ($p<0.1$).

Group	Response variable	Estimate	Statistic	P-value
C	BET	14.79	113.0	0.01
T	BET	10.96	148.5	0.02
C	logBET	0.99	118.5	0.00
T	logBET	0.40	128.0	0.09

Table 5: Pairwise tests of BET and logBET showing the difference between BET and logBET for C and T from T1 to T2. Estimate = difference between the two observations (T1 and T2) in a pair (Wilcoxon signed-rank test). C n=23; T n=23.

Thus, the tests demonstrate that there was a statistically significant improvement between T1 and T2 for both groups, even though the significance of logBET is a little weaker for T. There is also a counterintuitive suggestion of a greater improvement in C compared to T.

Taking into account the difference in the number of steps in the 3-AFC series, both groups had a significant improvement from T1 to T2 ($p < 0.01$), with an average reduction in concentration steps (Table 4) (Wilcoxon rank-sum test).

5.3.2 Difference between the control group and the training group

Despite a slightly lower BET and logBET registered for C in T1 (Table 4), there was no statistically significant difference on BET or logBET between C and T in either T1 or T2 ($p > 0.1$, Wilcoxon rank-sum test).

Both groups improved between the two tests, but improvement was not significantly different between C and T, as the Wilcoxon rank-sum test for both the difference, FC and LFC (Table 4), was not significant ($p>0.1$).

5.3.3 Comparison of high/less-sensitive and certain/uncertain assessors

The improvement from T1 to T2 was more substantial for the assessors with a high threshold in T1 (Figure 3), indicating that repeated exposure had more effect on the less-sensitive assessors.

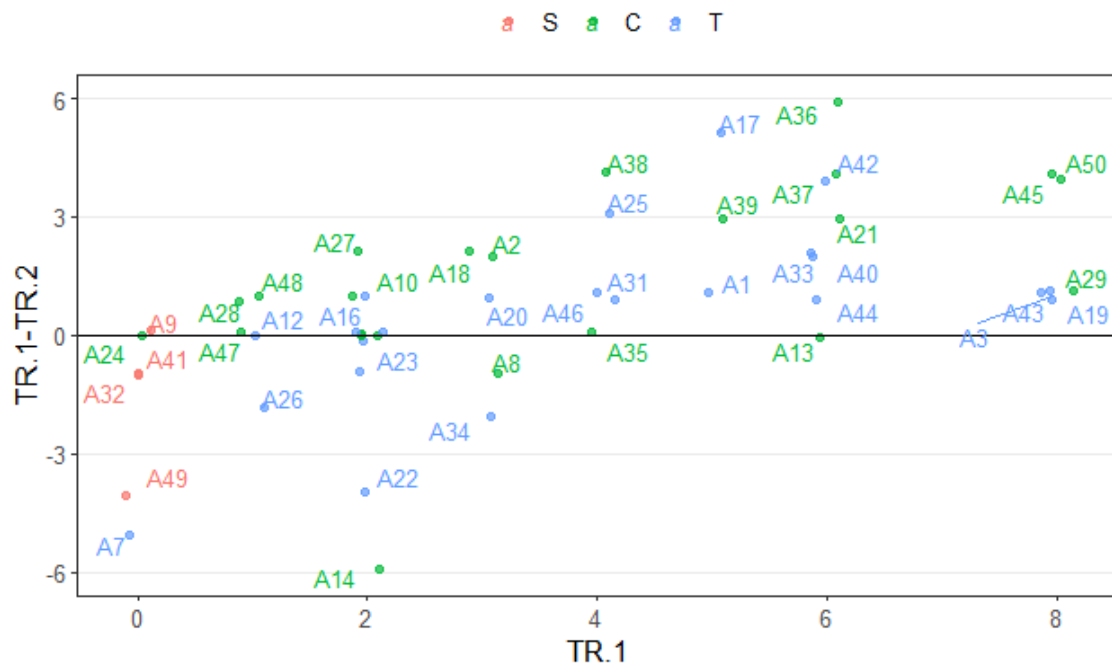


Figure 3: Performance of less-sensitive and high-sensitive assessors from T1 to T2. TR.1=Threshold T1 (last incorrect concentration before first correct), TR.1 – TR.2=Difference between threshold in T1 and T2. C n=23; T n=23; S n=4 (hyper-sensitive assessors). The assessors with the lowest sensitivity are on the right and have substantial improvements from T1 to T2, while the assessors with the highest sensitivity on the left are at the same level or slightly worse in T2.

To investigate whether the systematic training had a different effect for high/less-sensitive assessors, an ANOVA model was fitted with the effect of group (C, T) and

sensitivity group (L, H) and the interaction between them. The model was fitted with both LFC and rank (LFC), and results were comparable. Sensitivity group was highly significant ($p < 0.001$) with more improvement for L compared to H. When adjusting for sensitivity, there was in comparison a weaker significant difference between the groups (C, T) ($p = 0.05$ LFC, $p = 0.04$ rank LFC). However, the interaction between group and sensitivity group was not significant, meaning that the systematic training had no significantly different effect for L compared to H.

Before the pre-screening test, a total of 32 of the 50 assessors answered “yes” when asked whether they had detected mousiness in wine before. Two assessors gave negative answers to this question and the rest (16) were “uncertain”.

The group of assessors who had confirmed their certainty in detecting mousiness prior to the pre-screening test was compared to the assessors who were uncertain/had not previously encountered mousiness. The first group had a lower BET in T1 than the other group (Wilcoxon rank-sum test, $p = 0.03$).

5.4 Discussion Research Question 2 results

5.4.1 Induced sensitivity from repeated exposures

At panel level, the assessors achieved a lower detection threshold for ATHP following repeated exposures. The results were therefore in line with the research (2012) by Tempère, et al., who uncovered an induced sensitivity to diacetyl and linalool following a simple training procedure.

These are not the only examples where increased sensitivity has been documented for substances often related to a specific anosmia, the possible causes of the latter being thoroughly discussed in Wysocki, et al. (1989), who found that the ability to detect androstenone¹⁷ was induced in 10 out of 20 initially insensitive individuals who were systematically exposed to the compound. One of the researchers, who had initially been insensitive to androstenone, also experienced an induced sensitivity after months of interaction with the compound.

In the above research it was consequently suggested that there were three categories of human subjects: the truly anosmic, the inducible, and subjects who are naturally sensitive or have experienced incidental induction. Following the research by Wysocki, et al. (1989), both Cain & Schmidt (2002) and Dalton, et al. (2002) found that an induced sensitivity to the flavour of glutaraldehyde in drinking water, benzaldehyde and citralva, respectively, could be achieved through repeated exposures. The latter study found that the effect was limited to women, an effect which was not in line with the findings of, for example, Tempère, et al. (2012).

In this RP, the six assessors who did not detect C8 in T1 could initially qualify as “truly anosmic”. In T2, however, they were all able to detect C8 twice, making “the inducible” a more appropriate category. In fact, one of the assessors who did not detect C8 in either the pre-screening or in T1 (also confirming his complete lack of sensitivity after T1), appeared to “discover” the taint during the systematic training

¹⁷ A volatile steroid found in human perspiration, boar saliva, some pork products (in bacon, for example), truffles and celery (Wysocki, et al., 1989) (p. 7976).

programme, especially by nosing the vial in the tasting kit, which led to further retro-nasal detection. After completing the study, the same assessor also stated that during the training period, for the first time in his 20-year wine career he had encountered a wine he suspected was mousy, likely due to his increased awareness and/or detection ability as a consequence of the training. In fact, two of the pilot study participants had a similar experience of having acquired the ability to detect mousiness after being presented proper benchmarks for the taint.

In the pilot study, only one of the participants, despite the training, was still unable to detect the highest concentration in either the second sensitivity tasting or during a retest. When exposed to C8 used in the RP in a separate tasting to further investigate this participant, he could still not detect the strongest concentration of ATHP. Thus, this individual represented the only one of a total of 80 WPs who participated in either the pilot study or in the experiment in the RP who qualified as “truly anosmic”, even after repeated exposures.

5.4.2 Effect of systematic training versus repeated exposure

The systematic training of T did not lead to a more substantial reduction in detection threshold compared C, which perhaps could be expected due to a higher frequency of exposure. When comparing C and T, surprisingly, it was C that had a tendency to improve more.

In fact, in research by Tempère, et al. (2012), the increase in the WPs' sensitivity was only related to the compound (diacetyl or linalool) included in the systematic training programme, not the control compound, which also presented repeated exposure to the WPs of the study through repetition of the 3-AFC test. Also, in research by Wysocki, et al. (1989), the shift in sensitivity was unique to androstenone.

Contrary to these studies, less frequent exposures seemed to be enough to induce sensitivity to ATHP. While a single high intensity exposure (in this case, through the pre-screening) is not sufficient to train someone, multiple repeated exposures may be sufficient. However, a high level of exposure, as presented by the systematic training programme, seems unnecessary as there is no marginal gain versus less frequent repetitions. Nevertheless, following the study of the RP, as many as 17 assessors of T reported their impression of an increase in their own sensitivity following the systematic training. Thus, the effect of systematic training cannot be excluded.

Similar to studies by Tempere, et al. (2012) and Wysocki, et al. (1984) the decrease in the detection threshold was not uniform among all assessors of the RP.

Furthermore, the effect of repeated exposure was very uneven, ranging from small to substantial improvement.

5.4.3 Effects of awareness

Knowing what mousiness actually is, and having previous experience of the taint, may be of importance to detection ability and might at least partially explain the

differences reported in sensitivity (Cain & Schmidt, 2002). This was supported by the lower BET in T1 of the 32 assessors who responded affirmatively to detecting mousiness prior to the pre-screening. As WPs are perhaps less familiar with mousiness than, for example, 2,4,6-trichloroanisole (TCA), being certain of how to separate the taint from, for example, “brett character”, which shares some of the descriptors of mousiness (revealed in 5.5) such as animal and stable (Tempère, et al., 2014), appears to be important to detection.

In fact, 20 out of 23 assessors of T confirmed an increased certainty about their own ability to detect mousiness and 19 assessors confirmed an increased awareness after having completed the systematic training programme.

Mousiness is not mentioned in either WSET Level 1–3 or in the WSET Diploma materials (Master of Wine H. I. Hansen¹⁸ 2024, personal communication 23 April). Nor is it part of the one-day wine fault clinic in the Sommelier programme of the Culinary Academy in Norway. The fact that it is a relatively unknown wine fault among many WPs even though they have completed their formal wine education may therefore partially explain the lack of consensus regarding the taint.

¹⁸ Responsible for WSET levels 1-4 and the Sommelier education at the Culinary Academy in Norway

5.4.4 Potential effects of adaptation

In general, both C and T detected all concentrations more successfully in T2 compared to T1. However, both groups found it harder to detect C4 in T2 and T to detect C7 (Figure 1), indicating the effects of short-term adaptation.

Adaptation, however, is not only relevant to a threshold tasting. In fact, research by Dalton & Wysocki (1996) explored the long-term effects of odour exposure, finding a significant loss of sensitivity in the subjects of the study after two weeks of exposure to an odorant at home while an increase in sensitivity (although not significant) was registered for the control odorant. Before any exposure, the subjects in the above study completed two threshold tastings where the detection levels decreased significantly for both the control odorant and the adapting odorant from the first to the second test. Without long-term exposure, an increased sensitivity to both odorants across several tastings could therefore be expected.

These results may explain the tendency of increased improvement of C compared to T in the RP. With exposure to fewer, but still clear, benchmarks of mousiness, C did not seem to adapt as much as T.

The repeated exposure during the training programme for T was much less extensive than the research by Dalton & Wysocki's (1996), and therefore accounted for a potentially more limited long-term effect of adaptation. Contrary to the above study, the detection threshold of T in the RP actually decreased. Thus, it appears that the

training had a positive effect, although a less intense procedure might, in fact, have been even more successful in increasing sensitivity, since a partial effect of the training seemed to be a degree of adaptation in addition to sensitization (Dalton & Wysocki, 1996).

Thus, the most appropriate level of exposure for achieving the highest sensitivity possible is yet to be established and would be a good subject for further research. Whether a long-term exposure programme for truly anosmic subjects may induce sensitivity would also be worth exploring.

5.5 Research Question 3

What are wine professionals' descriptors of one of the mousy compounds, ATHP, in wine and how do these relate to established terms by previous literature?

A complete glossary can be found in Appendix 5.

5.5.1 Results and discussion

ATHP was first isolated in mousy wines in research by Strauss & Heresztyn (1984). Snowden, et al. (2006) summarizes research that dates from 1969 to 2002 (pp. 6467– 6468) regarding ATHP, APY and ETHP where ATHP is regarded as an important component of baked bread, crackers, taco shells, tortilla chips, popcorn and rice cakes, and is described as having a “cracker biscuit” and/or a “roast-

smelling” aroma. However, the latter two descriptions are also attributed to both APY and ETHP in their pure form.

Thus far, the sensory interactions between the three causative compounds of mousiness have not been explored (Snowdon, et al., 2006, p. 6466) and likely depend on their relative concentrations. In fact, when considering some of the more negative descriptors of mousiness outlined in 3.1.3 (regardless of the compound involved) such as “rodent urine”, “vomit” and “dirty mops”, it would be reasonable to assume that these occur at higher concentrations.

The properties of ETHP are less discussed in the literature. APY, on the other hand, is in previous research referenced in Snowdon, et al. (2006, p. 6467), explained as a key odorant present in wheat bread and particularly in more aromatic rice varieties such as Indian Basmati, as well as pandan leaves. Furthermore, it is regarded as an important compound in rice cakes, boiled trout, canned, frozen and fresh sweet corn, cooked lean beef, dried cured hams, lobster meat, green tea and even tiger urine, with descriptions such as “roasted and popcorn-like”, the latter two not unlike ATHP.

The assessors’ personal descriptors of C4 and C8 were used to evaluate whether these terms were similar to terms used in the literature in a qualitative context. Furthermore, they served to map the variation of descriptors of ATHP in general among WPs, and how they differed between a lower and a higher concentration. A total of 150 single descriptors were identified (Appendix 5). These were further

grouped in 30 main group descriptors by the researcher and revised by a colleague. The main group descriptors for C4 and C8 could therefore be compared (Figure 4).

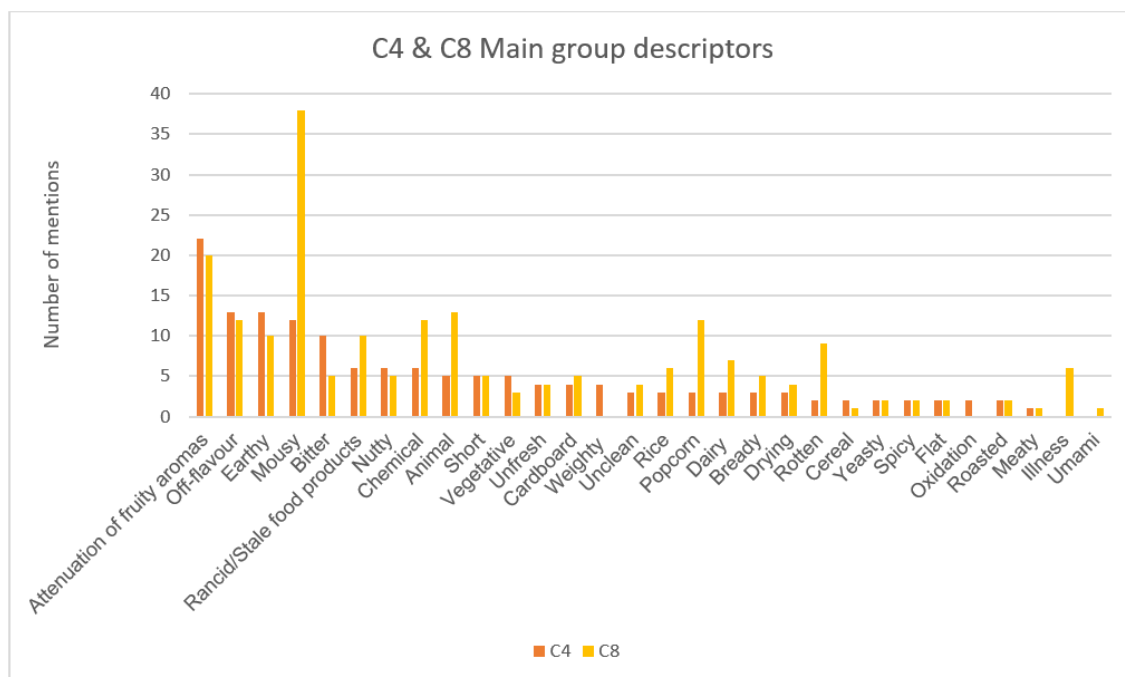


Figure 4: Main group descriptors of ATHP for C4 and C8 of the whole panel n=50.

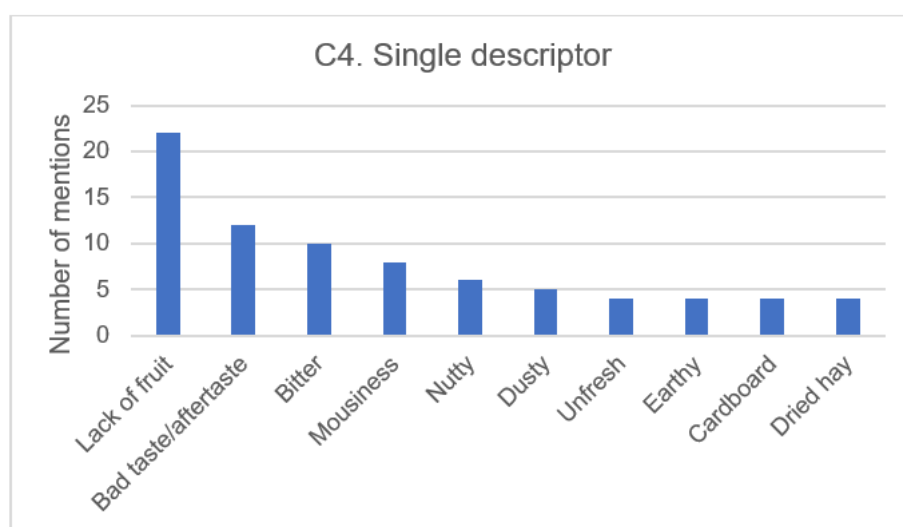


Figure 5: Top 10 single descriptors for C4 of the whole panel n=50.

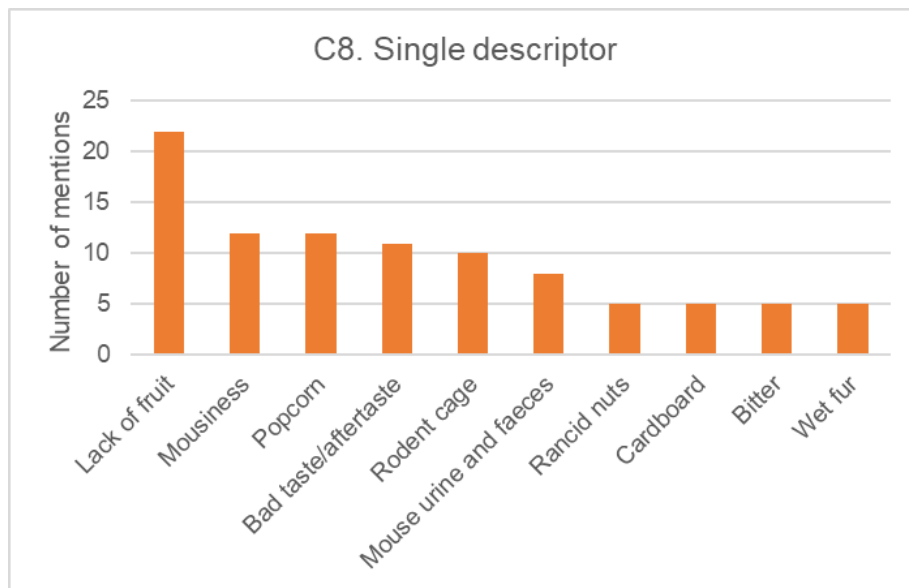


Figure 6: Top 10 single descriptors for C8 of the whole panel n=50.

The more numerous descriptors used for C8 compared to C4 (206 vs. 148 in total) indicated, first of all, the higher degree of complexity involved for a stronger concentration, but also indicates that C4 was perceived by fewer assessors. For C4, “attenuation of fruity aromas” (Figure 4) and “lack of fruit” (Figure 5) were by far the most common main group and single descriptor respectively. These indicated a vaguer impression of the taint and can be compared with, for example, TCA where a wine affected by cork taint at lower concentrations might be perceived as “less fruity” rather than clearly faulty. In comparison, the most common main group descriptor by far for C8 was “mousy” (Figure 4), with “mousiness”, “rodent cage” and “mouse urine and faeces” as the most frequently used single descriptors within the top main group descriptor (Figure 6).

Many of the main group descriptors were used for both concentrations. However, the frequency of the more negative descriptors such as “rancid/stale food products” and “rotten” was higher for C8. One of the most negative descriptors such as “illness” (including the single descriptors “nauseating,” “vomit” and “baby spit”) were in fact unique to C8. Thus, there would appear to be a relationship between the perception of retro-nasal odour quality and stimulus intensity for ATHP.

When comparing the assessors’ descriptors to terms previously used for ATHP, but also for ETHP and APY, in the literature outlined in both this section and in 3.1.3, they could all be found in the assessors’ vocabulary. Also, many of the descriptors used were the same or similar to terms referred to in the literature for APY in particular, indicating a sensory similarity between the different mousy compounds.

What is not known is whether the assessors’ descriptors, at least to some degree, were influenced by having read about mousiness prior to their participation in the experiment of the RP. While this cannot be ruled out, compared to the literature, some terms such as “roasted”, and “bready” (including “cracker”) were used by the assessors with an unexpectedly low frequency, and more emphasis was given to, for example, “attenuation of fruity aromas”, “bitter” and “chemical”, which are terms that are less discussed in the literature.

An interesting observation regarding descriptors related to mouthfeel was the wide range of single descriptors varying from “warming”, “weighty” and “viscous” to “astringent”, “thin” and “rustic”. This point in particular may indicate that the wide

variation in WPs' perception of mousiness goes beyond the retro-nasal smell itself, also affecting perception on the palate, especially in the aftertaste.

6. Conclusions

Mousy taint is perceived to be increasing and WPs struggle to detect it. Consumer risk is therefore growing. Although rapid quantification methods for ATHP, ETHP and APY are available for routine screening of wines, these do not replace sensory quality control. If repeated exposure to mousiness can induce sensitivity in WPs, we have a strong tool to address this challenge.

The results from the RP demonstrate that WPs' ability to detect mousiness in wine can be influenced by repeated exposures. ATHP can therefore be added to the wine-related compounds for which an induced sensitivity can be achieved.

Nevertheless, it is clear that an in-depth systematic training programme did not reduce detection thresholds any more than the simple repeated exposures that took place through repetition of the 3-AFC test. Therefore, we can conclude that all that is required for WPs to detect mousiness with more confidence is exposure to a limited number of clear benchmarks of ATHP.

Thus, a potential approach could be to include proper benchmarks of mousiness in wine education programmes such as WSET, Sommelier education and private programmes such as Campden BRI in the UK, in order to increase awareness, improve the detection ability and therefore also consensus among WPs regarding the taint.

Importantly, there was little evidence to support the widely-(mis)quoted figure of 25% for specific anosmia with respect to mousiness. Indeed, this RP suggests that a figure for true, specific anosmia may be much closer to zero. In other words, most subjects who cannot initially detect mousiness would be better described as “inducible subjects”, which makes them prime candidates for successful training, meaning there would be few limitations regarding the effectiveness of such a training programme in the wine trade at large.

The assessors’ descriptions of ATHP were extensive and both coincided with, and went beyond, what had previously been referred to in the literature for both ATHP, APY and ETHP, indicating a similarity in perception across the compounds. Retro-nasal odour quality appeared to depend on the level of concentration and negative descriptors were particularly relevant for the higher concentration. Thus, very weak concentrations of mousiness may not actually be a major issue to most individuals at all. As a follow-up to this study, measuring the consumer rejection threshold may uncover the concentration levels at which mousiness is in fact unpalatable from a mass-consumption perspective.

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7. Bibliography

ASTM E679, 2004. *Standard Practice for Determination of Odor and Taste Thresholds By a Forced-Choice Ascending Concentration Series Method of Limits*.
s.l.:s.n.

AWRI, 2011. *AWRI Annual Report*, Adelaide: Australian Wine Research Institute.

AWRI, u.d. *www.awri.com.au*. [Internet]

Available at:

https://www.awri.com.au/industry_support/winemaking_resources/sensory_assessment/recognition-of-wine-faults-and-taints/wine_faults/

Bartowsky, E. J. & Henschke, P. A., 1995. Malolactic fermentation and wine flavour. *Australian Grapegrower & Winemaker*, 378, p. 83–94.

Buttery, R. G., Ling, L. C., Juliano, B. O. & Turnbaugh, J. G., 1983. Cooked rice aroma and 2-acetyl-1-pyrroline. *Journal of Agricultural and Food Chemistry*, p. 823–826.

Cain, W. S. & Schmidt, R., 2002. Sensory detection of glutaraldehyde in drinking water - emergence of sensitivity and specific anosmia. *Chem Senses* 27, p. 425–433.

Costello, P. J., 1998. *Formation of mousy off-flavour in wine by lactic acid bacteria*. Adelaide: The University of Adelaide, Department for Horticulture, Viticulture and Oenology.

Costello, P. J., Lee, T. H. & Henschke, P. A., 2001. Ability of lactic acid bacteria to produce N-heterocycles. *Australian Journal of Grape and Wine Research*, p. 160–167.

Craig, J. T. & Heresztyn, T., 1984. 2-Ethyl-3,4,5,6-tetrahydropyridine - an assessment of its possible contribution to the mousy off-flavor of wines. *American Journal of Enology and Viticulture*, p. 46–48.

Dalton, P., Doolittle, N. & Breslin, P. A., 2002. Gender-specific introduction of enhanced sensitivity to odors. *Nat Neurosci*, p. 199–200.

Dalton, P. & Wysocki, C. J., 1996. The nature and duration of adaptation following long-term odor exposure. *Perception & Psychophysics*, 58, pp. 781–792.

EyeQuestion v.4.11.70, 2018. Elst, The Netherlands: s.n.

Feiring, A., 2022. Mouse of Ill Repute. *Noble Rot*, 2022(29), p. 99–101.

Goode, J., 2020. *Wine anorak*. [Internet]

Available at: <https://wineanorak.com/2020/04/24/the-latest-on-mouse-the-wine-taint-of-our-times/>

[Found 10 February 2022].

Goode, J., 2018. *Flawless*. 1 red. Oakland, California: University of California Press

Grapegrower & Winemaker, 2015. Ask the AWRI: Avoid mousy off-flavours.

Grapegrower & Winemaker, Issue 613, p. 50. Available at: [GGWM Feb2015 LR.pdf \(awri.com.au\)](#)

Grbin, P. R. et al., 1996. *Developments in the sensory, chemical and microbiological basis of mousy taint in wine*. Stockley, Adelaide, s.n., p. 57–61.

Grbin, P. R. & Henschke, P. A., 2000. Mousy off-flavour production in grape juice and wine by Dekkera and Brettanomyces yeasts. *Australian Journal of Grape Wine Research*, 6(3), p. 255–262.

Hayasaka, Y., 2019. Quantitative analysis of mousy off-flavour compound 2-acetyltetrahydropyridine in wine using liquid chromatography tandemmass spectrometry interfaced with atmospheric chemical ionisation. *Journal of Chromatography A*, 1588, p. 108–114.

Herderich, M., Costello, P. J., Grbin, P. R. & Henschke, P. A., 1995. Occurrence of 2-Acetyl-1-Pyrroline in Mousy Wines. *Natural Product Letters*, p. 129–132.

13301, ISO, 2018. *Sensory analysis-methodology-general guidance for measuring odour, flavour and taste detection thresholds by a three-alternative forced-choice*. s.l.:s.n.

ISO 8589, 2007. *Sensory analysis — General guidance for the design of test rooms*. s.l.:s.n.

ISO/IEC 17025, 2017. *Testing and calibration laboratories*. s.l.:s.n.

Kiyomichi, D. et al., 2023. Investigation into mousy off-flavor in wine using gas chromatography-mass. *Food Chemistry* 411, 15 June, pp. 1–8.

Larsen, M. J., Jensen, A. F., Madsen, D. M. & Pearce, E. F., 1999. Individual variations of pH, buffer capacity and concentrations of calcium and phosphate in unstimulated whole saliva. *Archives of Oral Biology*, p. 111–117.

Lawless, H. T., Antinone, M. J., Ledford, R. A. & Johnston, M., 1994. Olfactory responsiveness to diacetyl. *J. Sensory Stud* 9, p. 47–56.

Lawless, H. T. & Heymann, H., 2010. *Sensory evaluation of food, second edition*. s.l.:Springer.

Leske, P. A., Francis, I. L. & Hunt, D., 2013. *Sensory Evaluation*. s.l.:Australian Winemaking, eds N. Bulleid and V. Jiranek.

Macmillan, N. A. & Creelman, C. D., 1991. *Detection theory: a user's guide*. Cambridge University Press, New York, USA.

Massini, L. & Vuchot, P., 2015. Mieux cerner le défaut du « goût de souris". *Le vigneron des Côtes du Rhône et du Sud-Est*, n° 843, p. 26–27.

Müller-Thurgau, H. & Osterwalder, A., 1913. Die bakterien im Wein und Obstwein und die dadurch verursachten Veränderrungen. *Cent. für Bakt. ABT II*, pp. 141–142, 323–324.

O'Mahony, M., 1986. Sensory adaptation. *Journal of Sensory Studies*, 1, p. 237–257.

Oxford Companion to Wine, 2015. www.jancisrobinson.com. (Internet). Available at: <https://www.jancisrobinson.com/ocw/detail/mousy>

Parr, W. V., Heatherbell, D. & White, K. G., 2002. Demystifying wine expertise: Olfactory threshold, perceptual skill and semantic memory in expert and novice judges. *Chemical senses* 27, p. 747–755.

R Core Team, 2022. *R: A language and environment for statistical computing*, version 4.3.0., Vienna: s.n.

Snowdon, E. M., Bawyer, M. C., Grbin, P. R. & Bawyer, P. K., 2006. Mousy Off-Flavor: A Review. *Journal of Agricultural and Food Chemistry*, 8 August, p. 6465–6474.

Stevens, J. C., Cain, W. S. & Burke, R. J., 1988. Variability of olfactory thresholds. *Chemical senses*, 20, p. 643–653.

Strauss, C. R. & Heresztyn, T., 1984. 2-Acetyltetrahydropyridines – a cause of the “mousy” taint in wine. *Chem. Ind.*, pp. 109–110.

Tempère, S. et al., 2019. Comparison between standardized sensory methods used to evaluate the mousy off-flavor in red wine. *Oeno One*, Vol. 53 No. 2, 19 April, p. 1–10.

Tempère, S. et al., 2012. Explicit Sensory Training Improves the Olfactory Sensitivity of Wine Experts. *Chem Percept.*, 20 January, p. 205–213.

Tempère, S. et al., 2011. The Training Level of Experts Influences their Detection Thresholds for Key Wine Compounds. *Chem. Percept.*, 29 July, p. 99–115.

Tempère, S. et al., 2014. “Brett character” in wine: Is there a consensus among professional assessors? A perceptual and conceptual approach. *Food Quality and Preference* 34, p. 29–36.

Thudichum, J., 1894. *A Treatise on Wines*. London: George Bell & Sons.

Tucknott, O., 1974. *Taints in fermented juice products: mousy taint in cider*, s.l.: Annual Report Long Ashton Research Station: University of Bristol.

Venables, W. N. & Ripley, B. D., 2002. *Modern Applied Statistics with S*. 4 ed. New York: Springer.

Weerawatanakorn, M., Wu, J. C., Pan, M. H. & Ho, C. T., 2015. Reactivity and stability of selected flavor compounds. *Journal of Food and Drug Analysis*, 23(2), p. 176–190.

Wise, P. M., Bien, N. & Wysocki, C. J., 2008. Two Rapid Odor Threshold Methods Compared to a Modified Method of Constant Stimuli. *Chem. Percept.* 1, p. 16–23.

Woolf, S. J., 2017. *The Morning Claret*. [Internet]

Available at: <https://themorningclaret.com/2017/are-natural-winemakers-in-denial-about-mousiness/> [Found 29 March 2023].

Wysocki, C. J., Dorries, K. M. & Beauchamp, G. K., 1989. Ability to perceive androstenone can be acquired by ostensibly anosmic people. *Proc. Natl. Acad. Sci. USA*, 1 October, p. 7976–7978.

Appendix 1: Research Paper Proposal (RPP)

IMW Research Paper Proposal Submission Form			
Student ID	26040	Date of submission	May 24th 2024
RPP Version No	12	Name of Advisor	Tim Jackson MW
Note: RPPs must be submitted via your Advisor to the IMW			
Proposed Title			
Mousiness in wine: An investigation into the effects of repeated exposures and systematic training on wine professionals' sensitivity to mousiness			
Research Questions: Define the subject of your Research Paper and specify the specific research questions you plan to pursue. (No more than 200 words)			
Mousiness is referred to as «the wine fault of our times» by Jamie Goode (www.wineanorak.com, 2020), and several other authorities like Alice Feiring (personal communication, 2016) and Simon J. Woolf (www.themorningclaret.com, 2017) have expressed concern about the increasing number of mousy wines. No updated research exists on the actual occurrence of the taint on a larger scale, however, the perception of rising incidences is supported by various sources of research and studies (Snowdon, et al., 2006, AWRI, 2015, Tempère, et al., 2019, Kiyomichi, et al., 2023). Mousiness can render a wine unpalatable and hence represents a significant risk to a wine producer both in terms of costs and reputation. Moreover, there is still a lack of both consensus and awareness regarding the taint among wine professionals (Tempère, et al., 2019). Thus, discussions on the topic and differences in expert reviews can be both challenging and confusing as a wine may seem faultless to one person yet unpalatable to another. Several studies have found an induction of sensitivity in subjects after repeated exposure to a substance (Tempère, et al., 2012, Cain and Schmidt, 2002, Wysocki, et al., 1989). Thus far, no study has explored whether this also applies to mousiness. This research paper will therefore investigate the following: 1) What are wine professionals' detection levels of ATHP pre- and post-exposure? 2) Can wine professionals' sensitivity to mousiness be changed by repeated exposures and systematic training? 3) What are wine professionals' descriptors of one of the mousy compounds, ATHP, in wine and how do these relate to established terms by previous literature?			
Background and Context: Explain what is currently known about the topic and address why this topic requires/offers opportunities for further research. (No more than 200 words)			

Mousiness is perceived during retronasal evaluation (Bartowsky and Henschke, 1995), and is most commonly detected in a delayed fashion on the palate after the wine has been swallowed or expectorated. The off-flavour can rarely be perceived by sniffing as the mousy compounds are not sufficiently volatile at wine pH range (Grbin, et al., 1996). When the wine comes into contact with the higher pH range of a person's saliva, however, the taint can be detected retronasally. According to Massini and Vuchot (2015) the occurrence of mousiness in wines can be associated with significant decreases in the use of SO₂, increasing pH levels in wines, and the use of native microbiota in fermentations. The fault is therefore especially relevant for natural wines, which use minimal SO₂ and rely on native microbiota.

Mousiness can be of both microbial and chemical origin. Most strains of lactic acid bacteria and the yeast *Dekkera/Brettanomyces* can produce the character (AWRI, 2015).

Three *N*-heterocyclic molecules responsible for mousiness in wine have been identified, namely 2-acetyl-1-pyrroline (APY) (Herderich, et al., 1995), 2-acetyltetrahydropyridine (ATHP) (Strauss, et al., 1984) and 2-ethyltetrahydropyridine (ETHP) (Craig & Heresztyn, 1984). Of the three, ATHP is reported to be of major importance to the mousy off-flavour (Hayasaka, 2019). In Kiyomichi et al.'s recent research (2023) ATHP was detected in almost all of the wines suspected of mousiness in the study, which makes ATHP an interesting compound to investigate further in isolation in terms of detection thresholds. The ability to detect mousiness has been associated with a person's composition of saliva and pH on the tongue's surface (Grbin, et al., 1996). This may explain the findings from the Australian Wine Research Institute's (AWRI) fault tasting clinics, where, accumulating data from 2006 to 2011, only 75% of 238 wine professionals could detect the taint (AWRI Annual Report, 2011). This study is often referred to in articles about mousiness (although erroneously reported 70% instead of 75% which was the final result) (www.themorningclaret.com, 2017, www.wine-business-international.com, 2017, www.wineanorak.com, 2020) and even in wine literature such as The Oxford Companion to Wine. It should be noted that the AWRI did not conduct rigorous sensory investigations to collect the anosmia data. Moreover, the researchers couldn't determine the exact concentration of ATHP used in their experiment (Adrian Coulter, personal communication, 2022) so presumably it is difficult to say whether some participants were anosmic or simply less sensitive than others to whatever concentration was being provided.

In comparison, this study will investigate wine professionals' sensitivity to mousiness on an eight- step scale and whether induced sensitivity can be achieved by means of repeated exposures. The experimental design of this Research Paper is based on an in-depth pilot study carried out in 2022 on a panel of 30 wine professionals. As no previous research has investigated whether sensitivity to mousiness can be influenced by repeated exposures, insight from the pilot study has played an important role in defining the most appropriate methodology for answering the research questions of this RP.

Sources: Identify the nature of your source materials (official documents, books, articles, other studies, etc.) and give principle sources if appropriate. (No more than 150 words)

External provider of all statistical analysis including consultation on 3-AFC tests and experimental design:
NOFIMA (Norwegian Food Research Institute)

[Nofima - A leading institute for applied research within fisheries, aquaculture and food.](#)

Sensory scientist Josephine Skaret

Dr. scient and researcher (Statistics) Ingunn Berget

Lallemand: ¹⁹

Oenologist Marion Bastion and MScs Ann Dumont (Communication Manager)

Laboratorios Excell Ibérica S.L., Provider of concentrations of ATHP

Antonio Palacios

Advisors for the experimental design of the RP:

AWRI:

Senior Research Scientist Wes Pearson

Université de Bordeaux/INRAE/IPB, Unité de recherche Oenologie:

Maître de conférences à l'Institut des sciences de la vigne et du vin, Sophie Tempère

Books/ Articles on sensory training, 3-AFC Method and Statistics

- ASTM E679-19 Standard Practice for Determination of Odor and Taste Thresholds by a Forced-Choice Ascending Concentration Series Method of Limits¹
- Cain W. S and Schmidt R., 2002, Sensory detection of glutaraldehyde in drinking water - emergence of sensitivity and specific anosmia. *Chem Senses* 27(5): 425-433
- Dalton P., Doolittle N., Breslin P. A. S., 2002. Gender-specific induction of enhanced sensitivity to odors. *Nat Neurosci* 5(3):199-200
- ISO 13301-Second edition 2018-04. Sensory Analysis – Methodology – General guidance for measuring odour, flavour and taste detection thresholds by a three-alternative forced-choice (3-AFC) procedure
- ISO 6658- Third edition 2017-07. Sensory analysis- Methodology – General guidance
- Lawless H. T., and Heymann H., 2010. Sensory evaluation of food Principles and practices. Second edition, Springer, p. 129-133
- Leske P. A., Francis I. L., Hunt D., 2013. Sensory Evaluation. Australian Winemaking, eds N. Bulleid and V. Jiranek, online www.trivinum.com.au/awm/SEN
- *R Core Team (2022). R: A language and environment for statistical ## computing. R Foundation for Statistical Computing, Vienna, Austria. ## URL <https://www.R-project.org/>.*
- Tempère S., Cuzange E., Bougeant J. C., de Revel G., Sicard G., 2012. Explicit Sensory Training Improves the Olfactory Sensitivity of Wine Experts. *Chemosensory perception* 2012, 5:205-213
- Tempère S., Cuzange E., Malak J., Bougeant J. C., de Revel G., Sicard G., 2011. The training level of experts influences their detection thresholds for key wine compounds. *Chemosensory Perception* 2011, 4:99-115.
- Wise, P. M., Bien, N. & Wysocki, C. J., 2008. Two Rapid Odor Threshold Methods Compared to a Modified Method of Constant Stimuli. *Chem. Percept.* 1, pp. 16–23.
- Wysocki, C. J., Dorries, K. M. & Beauchamp, G. K., 1989. Ability to perceive androstenone can be acquired by ostensibly anosmic people. *Proc. Natl. Acad. Sci. USA*, 1 October, pp. 7976–7978.

¹⁹ Sponsor of the concentrations of ATHP used in Research Paper

Other articles/literature on mousiness for context:

- Costello P. J., Lee T. H., Henschke P. A., 2001. Ability of lactic acid bacteria to produce N-heterocycles causing mousy off-flavour in wine. Australian Journal of grape and wine research 7, 160-167.
- Grbin, P.R., Costello, P.J., Herderich, M., Markides, A.J., Henschke, P.A. and Lee, T.H., 1996. Developments in the sensory, chemical and microbiological basis of mousy taint in wine. Proceedings of the 9th Australian Wine Industry Technical Conference, Winetitles, Adelaide, Pag. 57-61.
- Hayasaka Y., 2019. Quantitative analysis of mousy off-flavour compound 2-acetyltetrahydropyridine in wine using liquid chromatography tandem mass spectrometry interfaced with atmospheric chemical ionisation. Journal of Chromatography A, 1588 (2019), pp. 108–114.
- Kiyomichi D., Frank C., Moulis P., Riquier L., Ballestra P., Marchand S., Tempère S., de Revel G., 2023. Investigation into mousy off-flavor in wine using gas chromatography-mass spectrometry with stir bar sorptive extraction. Food Chemistry 411 (2023), 135454
- Massini L., Vuchot P., 2015. Mieux cerner le défaut du “gout de souris”. Le vigneron des Côtes du Rhône et du Sud-Est n. 843-26-27.
- Snowdon E. M., Bowyer M.C., Grbin P.R. and Bowyer P.K., 2006. Mousy off-flavor: a review. Journal of Agricultural and Food Chemistry, 54(18), 6465–6474.
- Tempère S., Chatelet B., de Revel G., Dufoir M., Denat M., Ramonet P. Y., Marchand S., Sadoudi M., Richard N., Lucas P., Miot-Sertier C., Claisse O., Riquier L., Perello M. C., Ballestra P., 2019. Comparison between standardized sensory methods used to evaluate the mousy off-flavour in red wine. OENO One, 53(2).
- Goode, J., 2020, www.wineanorak.com [The latest on mouse, the wine taint of our times – wineanorak.com](http://www.wineanorak.com)
- Mccarrol j., 2022, Punch Drink: [Mousiness Is Natural Wine's Biggest Flaw | PUNCH \(punchdrink.com\)](http://punchdrink.com)
- Williams P., Oxford Companion to Wine [mousy | JancisRobinson.com](http://JancisRobinson.com). Accessed 04/12/2023.
- Woolf S. J., 2017, www.themorningclaret.com [Are natural winemakers in denial about mousiness? | The Morning Claret](http://www.themorningclaret.com)
- Woolf S. J., 2017, www.wine-business-international.com, [The mousiness problem | Meiningers Wine Business International \(wine-business-international.com\)](http://www.wine-business-international.com)
- **AWRI research on mousiness** – [GGWM Feb2015 LR.pdf \(awri.com.au\)](http://awri.com.au), 2015 (this fact sheet is not up to date, however it refers to the 30 % of anosmic subjects reported in mentioned literature/articles above)
- **AWRI Annual Report, 2011** (see attached file). Contains the *actual* number of participants of the study and the final correct percentage of detection among the participants (75 % and not 70 %)

Research Methodology: Please detail how you will identify and gather the material or information necessary to answer the research question(s) and discuss what techniques you will use to analyse this information. (No more than 500 words)

The experimental design is based on previous research, books on sensory evaluation and methods, ISO and ASTM standards, findings from the pilot study and further discussions with sensory scientists, researchers and statisticians defined under “Sources”.

The overall objective of the research is to measure *changes in sensitivity* to one of the mousy compounds, ATHP, in the specific population of “wine professionals” through repeated exposures. Changes in detection thresholds are used as a *tool* to evaluate the potential impact of these exposures. To investigate whether a potential change is due to repetitions of the 3-AFC test during “Test 1” and “Test 2” (ISO 13301-18) described below or systematic training procedure, the participants will be divided into a training group and a control group. The details of the methodology are described below.

The experiment will consist of three phases: two test sessions separated by a training period of approx. 2 weeks.

Phase 1: Test session 1

Objective: Measure the detection threshold of mousiness (ATHP) of each participant

- **Participants of the panel:** The panel will consist of wine professionals from different branches of the Norwegian wine trade: Employees of the Norwegian Monopoly (retail), wine importers, educators and wine journalists. The participants will all have completed minimum WSET 3/Sommelier degree and currently work in the wine trade with wine tasting as part of their work. They will know in advance that they will taste for mousiness and their participation is based on their own motivation of learning.
- **Panel size:** The intention is to recruit around 40 participants, however a minimum of 30 participants will be accepted as the final panel size. Twenty-five participants are considered sufficient for threshold tastings by Lawless and Heymann (2010, p. 130), while Leske, et al., 2013 suggests a minimum of 15-20 panelists for difference testing. The aim of the research is not to obtain the absolute detection threshold value for ATHP, but to evaluate if it is possible to induce an increased sensitivity in the participants by means of repeated exposures. For the same purposes Tempère, et al.’s study (2012) had 16 participants. The suggested minimum of participants for this research will therefore allow for some unforeseen drop out during the experiment.
- **Pre-screening of panelists:** The panel will consist of both participants who prior to the 3-AFC tests confirm their detection of mousiness based on exposure to the highest concentration of ATHP used in the experiment, and participants who fail to detect it. All participants will therefore have tasted the same benchmark for mousiness prior to the tastings, contrary to the pilot study.
- **Randomized groups for training and control:** The participants will be equally divided into two different groups. One group will take part in an individual training procedure in between the two 3-AFC test sessions, while the other group will serve as a control group and will not take part in the training procedure. The participants who do not detect the highest concentration of ATHP during the pre-screening test and/or the first sensitivity tasting measurement will be equally randomized in the training and control group.
- **Location:** The tastings will be organized in the Norwegian Wine Monopoly’s seven ISO – certified isolated tasting booths with an average consistent room temperature of 19–21°C.
- **Performance of 3-AFC tests by participants:** All participants will be exposed to eight 3-alternative forced choice tests (3-AFC, ISO 13301-2018) in order of ascending

concentrations of one of the mousy compounds, ATHP. The participants are asked to taste and identify the mousy sample in each of the eight 3-AFC tests, giving their answers in the sensory software EyeQuestion. If they are uncertain, they are told to guess (Lawless and Heymann, 2010, p. 129). The participants can taste each sample only once to avoid fatigue. They will not be informed of the results during the tasting. During tasting, the participants are asked to expectorate the wine after 4-8 seconds. A minimum break of 2 minutes between each wine is required to avoid residual mousiness to affect a following clean sample. This prolonged recovery time is based on the findings of the pilot study, uncovering the strong adaptation effect. The participants are also encouraged to wait even longer between samples and/or trials if necessary, as the highest concentrations may require an even longer recovery time depending on individual sensitivities. The participants can rinse their palate with neutral table water in between the samples. The participants will also have small pieces of bread/neutral biscuits to eat in between wines if in need to neutralize their palate after a mousy example. The participants must complete all eight 3-AFC tests to minimize the false positive rate.

- **Concentrations:** The concentrations of ATHP are provided by Laboratorios Excell Ibérica S.L. The concentration range will be based on the pilot study, previous concentrations found in wines (Kiyomichi et al., 2023, Grbin et al., 1996) and a new pilot testing of the selected concentrations to ensure that they cover a various range of thresholds. The concentrations of ATHP will be provided in solutions in small vials. 1 vial shall be added to one 75 cl bottle of wine 2 hours before the tasting (the wines will be spiked in the Wine Monopoly's premises in Oslo where the tastings are held). Whereas the pilot study combined ATHP and APY at each concentration step with no consistent dilution factor, a consistent dilution factor between the eight concentrations of the 3-AFC test will be applied in order to 1) calculate the Best Estimate of individual threshold (BET) for the subsequent analysis and 2) avoid a potential matrix effect between the compounds.
- **BET – measurements and criteria for correct answers:** Lawless and Heymann (2010, p. 131) defines BET as “the geometric mean of the first correct trial with all subsequent trials correct and the previous trial incorrect”. If all the answers of a participant are correct throughout the entire series it is the geometric mean of the lowest concentration and the next hypothetical lower concentration. At the top it is the geometric mean of the highest concentration tested and the next hypothetical higher concentration. Previous research such as Tempère, et al. (2011, 2012, 2019) and Wise, et al. (2008) on the other hand, have applied other criteria for a “correct answer” such as the geometrical mean between the lower concentration *in three consecutive answers* and the previous concentration. The quality of the data achieved for this experiment will determine the most suitable approach in this respect and will be explained and justified in the RP.

Phase 2 : Approx. 2 weeks of individual sensory training period for half of the participants of the panel

After completing the first test session, the participants assigned to the training group are provided with the following to last over the individual training period:

- **Materials:** Small 2 oz glass bottles with wine spiked with the highest concentration of ATHP used in the 3-AFC tests and a set of identical bottles of a lower concentration used in the same tests. A small vial containing residues of the supraliminal concentration to be

nosed during the training period as it gives a clear aroma of the taint even without tasting it (due to the pyridines dissolved in ethanol).

- **Training period:** Results from the pilot study revealed that some participants felt more insensitive to mousiness following the intensive training period. For the next experiment participants will therefore be asked to alternate training with the higher and lower concentration of ATHP during the 2 weeks. Each training session will last for max. five minutes and the participants will log on into EyeQuestion by means of a QR code on the samples to track their commitment to the training. Previous sensory training research on other compounds have included training periods running from five to sixty days with different intensities (Tempère et al., 2012, Cain and Schmidt, 2002, Dalton et al., 2002). The 2 week-period proposed for this research sits in the middle of this range and appropriate length and intensity will be based on the pilot study.

Phase 3: Test session 2

Objective 1: Track any changes to the panel's sensitivity to mousiness.

Objective 2: Compare any differences in detection ability between the group of participants subject to training and the control group

Objective 3: Obtain descriptors of ATHP from the participants to answer RQ3

- The second tasting session will be organized in the same location and under the same circumstances as test session 1.
- The participants will be exposed to the exact same eight ascending 3-AFC tests as in the first test session.
- After completion of the 3-AFC test and a short break, the participants are asked to describe a lower and a higher concentration of ATHP by means of two separate glasses

Base wine: The same base wine will be used for the pre-screening, both test sessions and the training period (Laroche Chardonnay L 2022).

Statistical methods:

Linear regression with the proportion of assessors detecting the mousy-sample as the dependent variable and the concentration levels (on log scale) as independent variable, will be applied to test if there is i) a significant difference at the panel level for the two rounds of test, and ii) if the difference between the two rounds differs between the training and the control group. In addition, individual BET-estimates for the two test rounds will be compared for the training and control group using parametric and non-parametric tests. In all analyses the significance level will be set to $\alpha=0.05$. All data analysis will be performed with R (R-project²⁰).

Potential to Contribute to the Body of Knowledge on Wine: Explain how this Research Paper will add to the current body of knowledge on this subject. (No more than 150 words)

²⁰ See "sources"

Taints and off-flavours represent a major concern in the wine industry and mousiness is no exception to the rule. Up to now, most research on the subject has been devoted to understanding the causes of mousiness and recently about quantification of the compounds involved (Kiyomichi, et al., 2023, Hayasaka, 2019). The most referenced research to date on detection rates of mousiness (AWRI, 2011), however, concluded that 25 % of wine professionals could not detect the taint. Although this result is commonly explained by tasters' different composition of saliva and pH on the tongue's surface, no research has so far explored whether repeated exposures to one of the compounds responsible for the taint may influence wine professionals' detection ability. If so, repeated exposures may potentially represent a tool to both reduce the lack of consensus among wine professionals and increase their abilities for sensory quality control in wine under normal wine tasting conditions.

Proposed Time Schedule/Programme: This section should layout the time schedule for the research, analysis and write-up of the Research Paper and should indicate approximate dates with key deliverables. *Dates of submission to both Advisors and the IMW must be those specified by the IMW.*

May

- **May 31st: Proof reading of RP**

June

- **June 6th: Deadline any amendments to RPP or RP title**
- **June 20th: Final submission of paper**

Appendix 2: List of Abbreviations

Abbreviation	Meaning
3-AFC	Three-alternative forced-choice
AIC	Akaike information criterion
ANOVA	Analysis of Variance
APY	2-acetyl-1-pyrroline
ATHP	2-acetyltetrahydropyridine
AWRI	Australian Wine Research Institute
BET	Best Estimate Threshold
C	Control group
C1–C8	Range of concentration levels
ETHP	2-ethyltetrahydropyridine
FC	Fold change
H	High-sensitive assessor
L	Less-sensitive assessor
LFC	Log fold change
PC	Probability of correct
PD	Probability of detection
RP	Research paper
S	Hyper-sensitive assessor
T	Training group
T1	First sensitivity tasting
T2	Second sensitivity tasting
WP	Wine professional

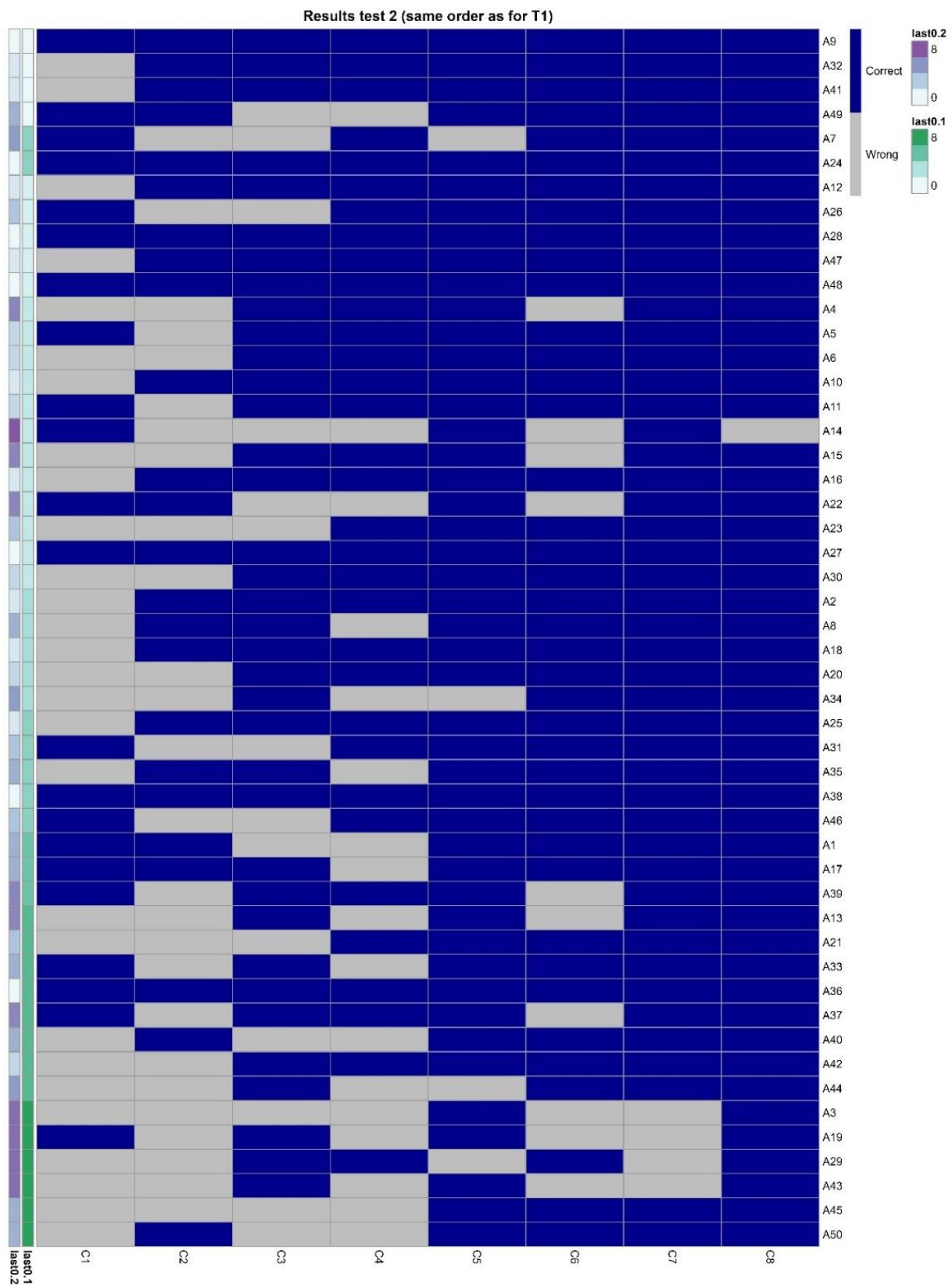
Appendix 3: Assessors' individual detection thresholds T1, T2

BET and thresholds from T1 and T2. Group: C= Control group, T=Training group, S=Hyper-sensitive. Certainty of having detected mousiness prior to pre-screening Y=Yes, N=No, U=Uncertain, TR=Threshold (last incorrect concentration before first correct of three correct answers in a row), last0=Last incorrect concentration in the whole series (different from TR if one incorrect answer after 3 correct answers). BET 1= BET estimate T1, BET 2= BET estimate T2. In T1, A19 and A29 had several incorrect answers after three correct answers in a row. These two assessors therefore have the highest BET value.

Assessor	Group	Certainty mousiness	Test1			Test 2		
			TR1	last0	BET1	TR2	last0	BET2
A1	T	Y	5	5	12.353	4	4	4.941
A2	C	Y	3	3	1.976	1	1	0.316
A3	T	U	8	8	193.010	7	7	77.204
A4	T	Y	2	2	0.791	2	6	0.791
A5	C	Y	2	2	0.791	2	2	0.791
A6	T	Y	2	2	0.791	2	2	0.791
A7	T	Y	0	4	0.126	5	5	12.353
A8	C	Y	3	3	1.976	4	4	4.941
A9	S	Y	0	0	0.126	0	0	0.126
A10	C	Y	2	2	0.791	1	1	0.316
A11	T	Y	2	2	0.791	2	2	0.791
A12	T	Y	1	1	0.316	1	1	0.316
A13	C	U	6	6	30.882	6	6	30.882
A14	C	Y	2	2	0.791	8	8	193.010
A15	C	U	2	2	0.791	2	6	0.791
A16	T	Y	2	2	0.791	1	1	0.316
A17	T	U	5	5	12.353	0	4	0.126
A18	C	U	3	3	1.976	1	1	0.316
A19	T	Y	8	8	193.010	7	7	77.204
A20	T	U	3	3	1.976	2	2	0.791
A21	C	Y	6	6	30.882	3	3	1.976
A22	T	U	2	2	0.791	6	6	30.882
A23	T	U	2	2	0.791	3	3	1.976
A24	C	Y	0	4	0.126	0	0	0.126
A25	T	Y	4	4	4.941	1	1	0.316
A26	T	Y	1	1	0.316	3	3	1.976
A27	C	Y	2	2	0.791	0	0	0.126
A28	C	U	1	1	0.316	0	0	0.126
A29	C	Y	8	8	193.010	7	7	77.204
A30	C	Y	2	2	0.791	2	2	0.791
A31	T	Y	4	4	4.941	3	3	1.976
A32	S	Y	0	0	0.126	1	1	0.316
A33	T	U	6	6	30.882	4	4	4.941
A34	T	Y	3	3	1.976	5	5	12.353
A35	C	Y	4	4	4.941	4	4	4.941
A36	C	N	6	6	30.882	0	0	0.126
A37	C	Y	6	6	30.882	2	6	0.791
A38	C	U	4	4	4.941	0	0	0.126
A39	C	Y	5	5	12.353	2	6	0.791
A40	T	U	6	6	30.882	4	4	4.941
A41	S	Y	0	0	0.126	1	1	0.316
A42	T	U	6	6	30.882	2	2	0.791
A43	T	U	8	8	193.010	7	7	77.204
A44	T	U	6	6	30.882	5	5	12.353
A45	C	Y	8	8	193.010	4	4	4.941
A46	T	N	4	4	4.941	3	3	1.976
A47	C	Y	1	1	0.316	1	1	0.316
A48	C	Y	1	1	0.316	0	0	0.126
A49	S	U	0	0	0.126	4	4	4.941
A50	C	Y	8	8	193.010	4	4	4.941

Heatmap T2

Same type of figure as for Heatmap T1, but with data from T2, grouped in the same order as T1

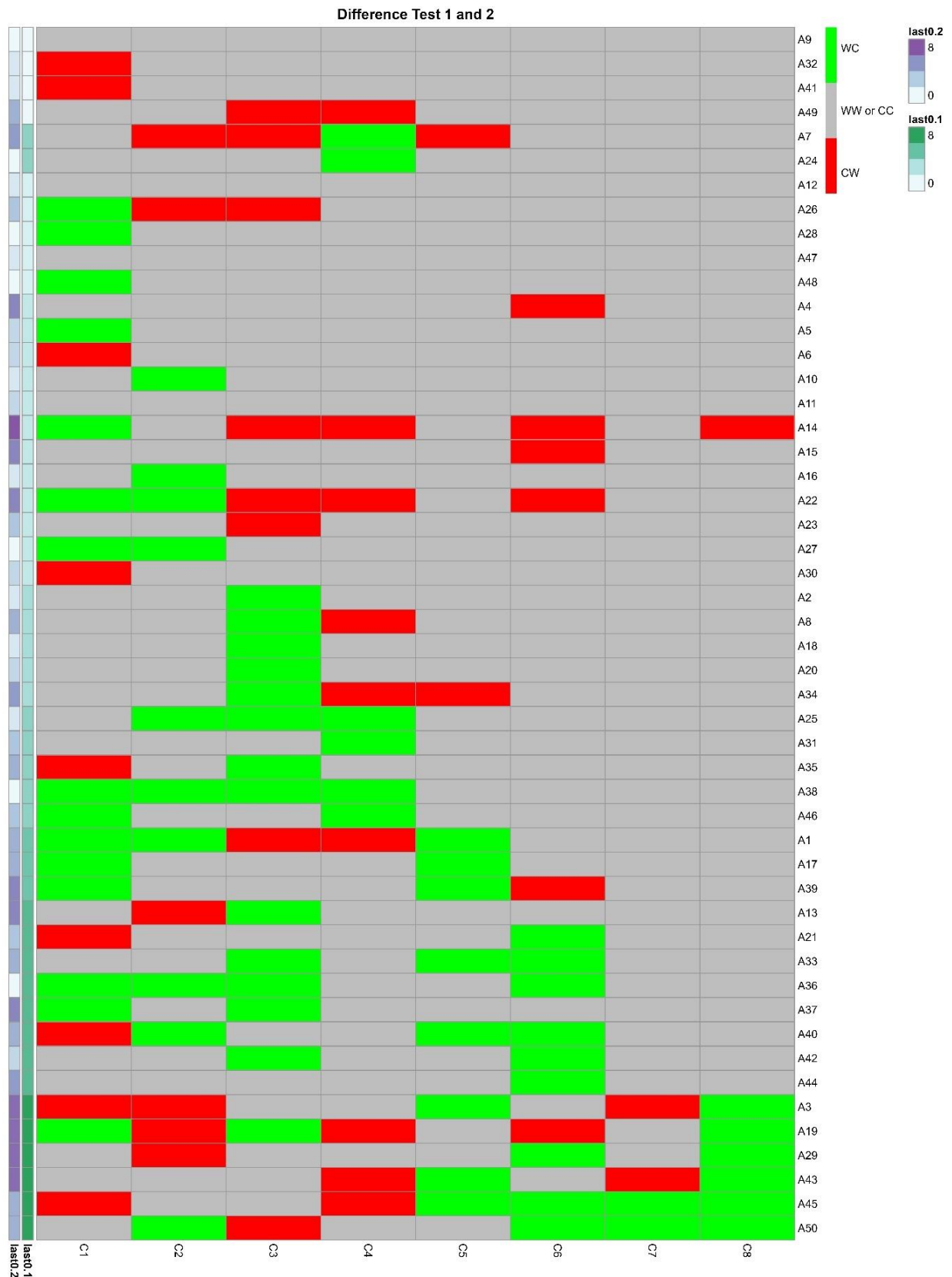


Heatmap T2-T1

Red= Correct (C) T1/Wrong (W) T2 (CW)

Green= Wrong T1/Correct T2 (WC)

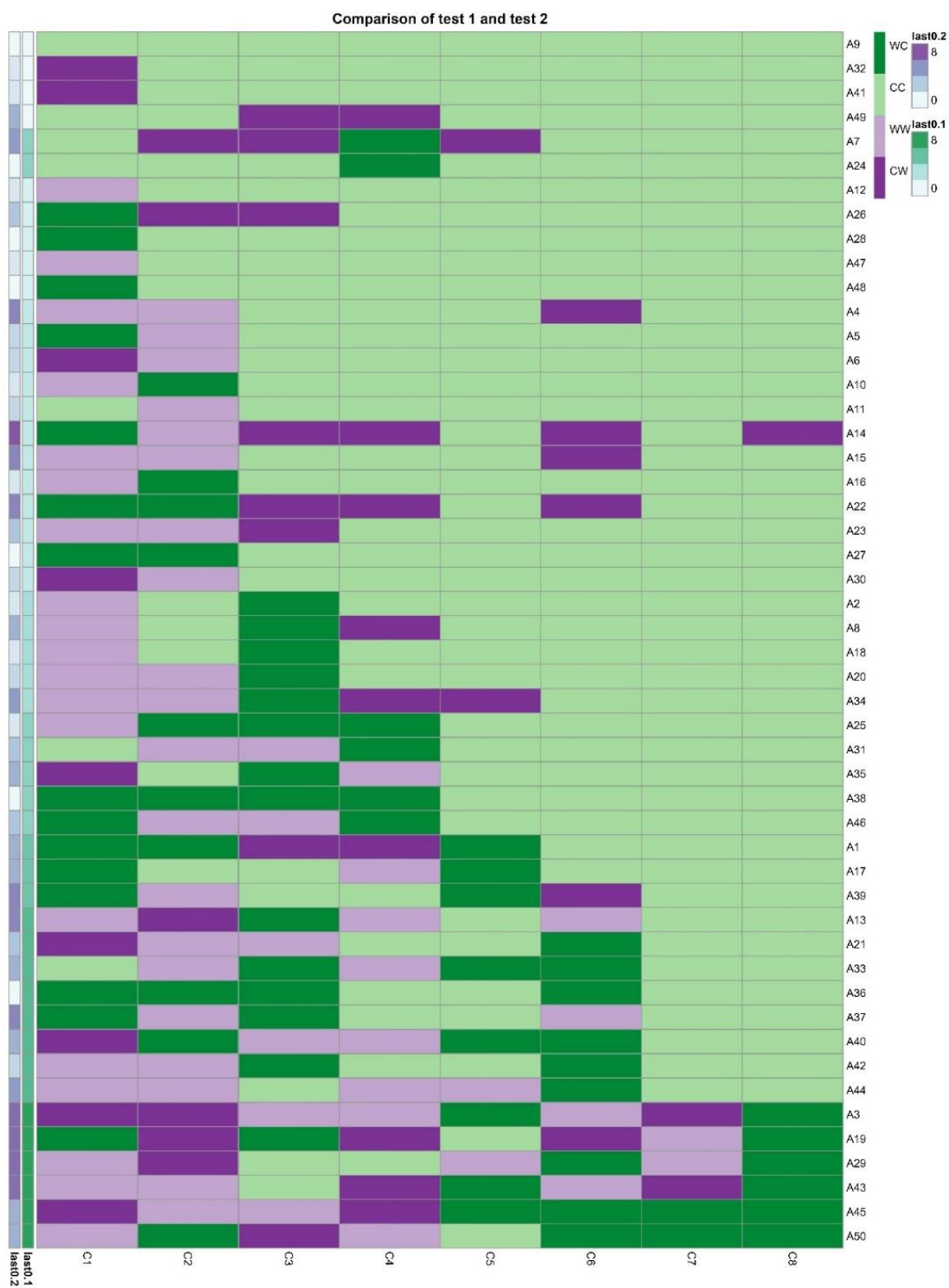
Grey= Same answer in T1 and T2 (WW or CC)



Heatmap comparison T1, T2

Green: Either correct in both T1 and T2 (CC) or wrong in T1 and correct in T2 (dark green) (WC)

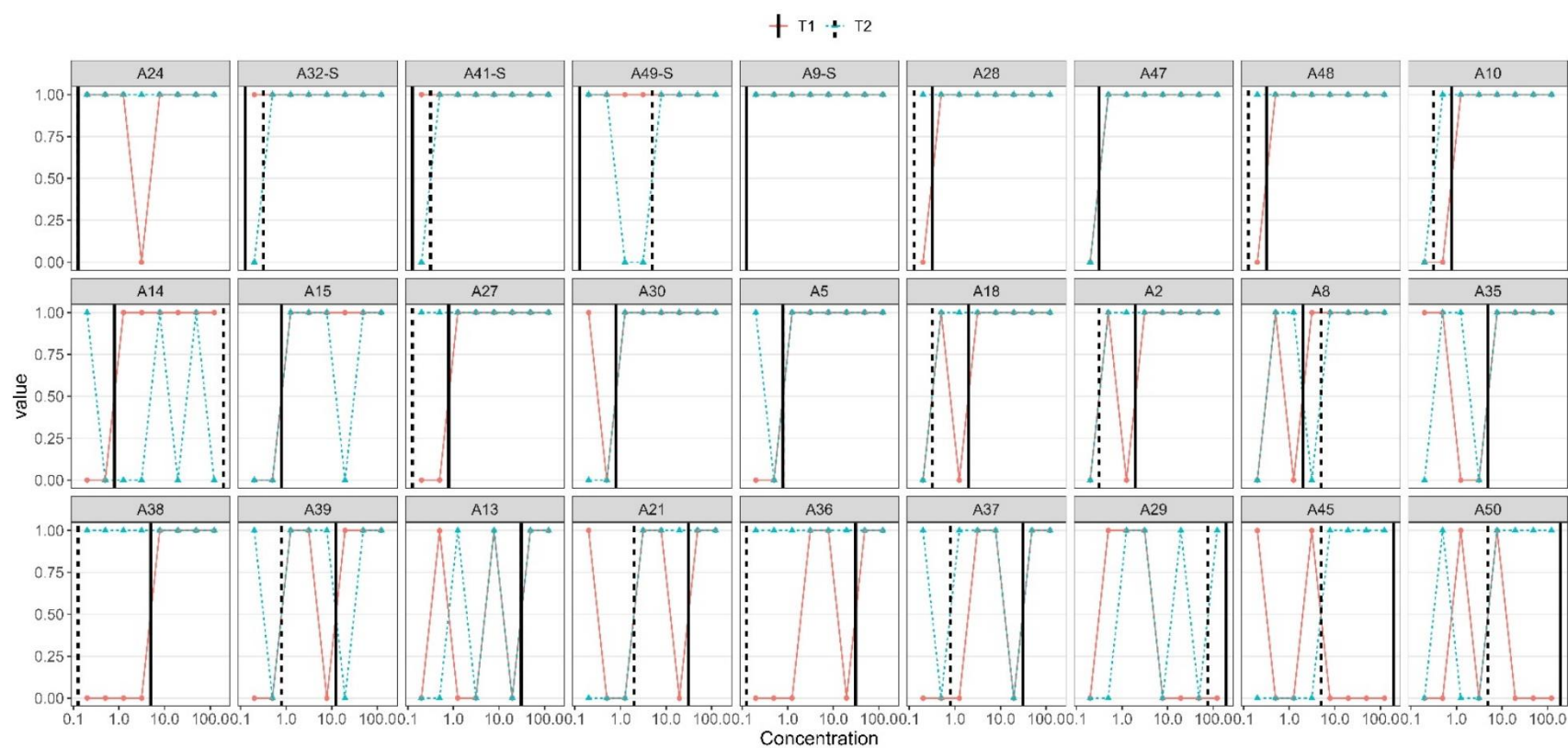
Purple: Either wrong in both T1 and T2 (WW) or correct in T1 and wrong in T2 (dark purple) (CW)



Individual curves T1 and T2 together, incl. BET

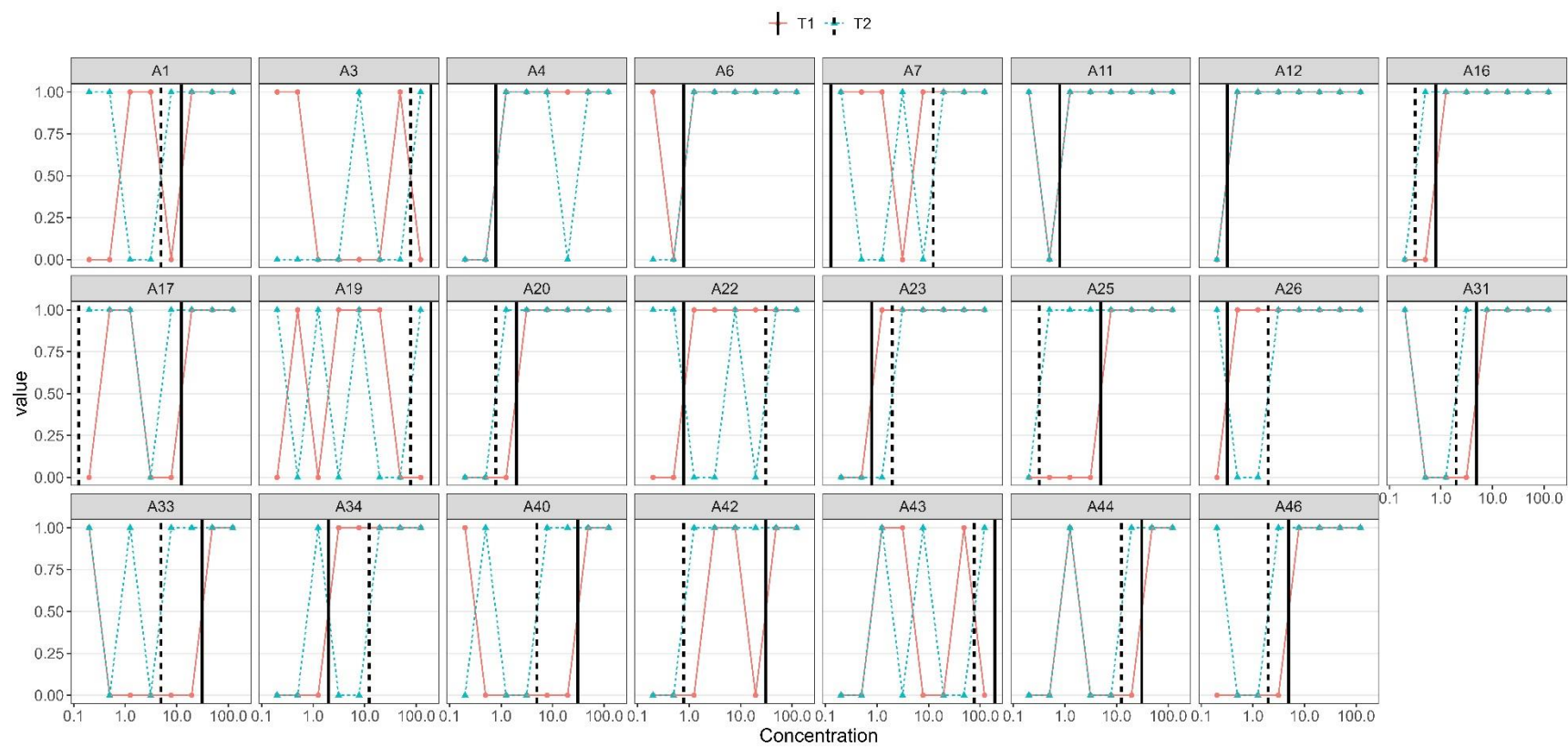
Vertical line: BET T1, vertical dotted line: BET 2. If one line only: Same BET in T1 and T2.

Results of Control group (C) and 4 hyper-sensitive assessors (S):



Individual curves T1 and T2 together, incl. BET

Results of training group (T):



Appendix 5: Glossary

Complete glossary: Main group and single descriptors. Main group descriptors sorted according to frequency for C4.

Main group descriptor	C4	C8	Single descriptor	Single descriptor	Single descriptor	Single descriptor	Single descriptor
Attenuation of fruity aromas	22	20	Lack of fruit	Unfocused	Attenuation of nuances	Attenuation of aromatics	Complete lack of fruit
Off-flavour	13	12	bad taste in the mouth	Bad aftertaste	Unpleasant aftertaste	"Off"	Funky
			Negative complexity				
Earthy	13	10	Dusty	Smell of old locker	Mushroom	Cellar	Wet dark soil
			Bad mushrooms	Aftertaste of fresh mushrooms	Damp swamp	Mouldy	Mouldy bread
			Dry dark soil				
Mousy	12	38	Mouse	Rodent cage	Mouse cage	Mouse urine and faeces	Hamster cage
			Dirty wood shavings	Dead rodent	Old attic with mice	Guinea pig cage	
Bitter	10	5	Bitter almond	Bitter finish	Phenolic		
Rancid/Stale food products	6	10	Rancid nuts	Stale biscuits	Rancid cereal	Stale popcorn	Rancid butter
			Rancid corn tortilla	Rancid fat			
Nutty	6	5	Almond skin				
Chemical	6	12	Sulfur	Cooked cabbage	Pungent	Volatile acidity	Nail polish
			Aspartane	Synthetic	Bad jelly belly flavours	Glue	Medicinal
			Sweet vinegar	Matchstick	Metallic	Ammonia	Soap
Animal	5	13	Wet fur	Furry	Wet leather	Zoo	Circus with exotic animals
			Dry cat food	Fish skin	Stable	Sweat	
Short	5	5	Scathing mouthfeel	Thin			
Vegetative	5	3	Dried hay	Hay	Genmai green tea		
Unfresh	4	4	Lack of acidity	Attenuation of acidity			
Cardboard	4	5	Wet cardboard				
Weighty	4	0	Viscous	Alcoholic	Warning		
Unclean	3	4	Dirty kitchen cloth	Dirty socks			
Rice	3	6	Boiled rice	Overcooked rice			
Popcorn	3	12	Butter popcorn				
Dairy	3	7	Lactic	Butter	Milk	Yoghurt	Cheese
			Cheese doodles	Spoiled dairy			
Bready	3	5	Crackers	Bread crust	Brioche	Toast	Sourdough breath
			Sourdough bakeries				
Drying	3	4	Astringent	Tannic	Rustique		
Rotten	2	9	Rotten hay, silage	Rotten eggs	Rotten meat	Rotten vegetables	
Cereal	2	1	Grainy	Floury			
Yeasty	2	2	Uncompleted fermentation	Bread dough	Yeast		
Spicy	2	2	Barbecue spice	Vanilla			
Flat	2	2	Flabby	Lack of energy			
Oxidation	2	0	Old lemon juice	Oxidized apple			
Meaty	1	1	Liverpaté				
Roasted	2	2	Toast	Burned toast	Roasted nuts		
Illness	0	6	Nauseating	Vomit	Baby spit		
Umami	0	1					
Total	148	206					

Main group descriptors C4 and C8.

N=Number of times descriptor used by assessors.

C4			C8		
Rank	Main group descriptor	N.	Rank	Main group descriptor	N.
1	Attenuation of fruity aromas	22	1	Mousy	38
2	Off-flavour	13	2	Attenuation of fruity aromas	20
3	Earthy	13	3	Animal	13
4	Mousy	12	4	Off-flavour	12
5	Bitter	10	5	Chemical	12
6	Rancid/Stale food products	6	6	Popcorn	12
7	Nutty	6	7	Earthy	10
8	Chemical	6	8	Rancid/Stale food products	10
9	Animal	5	9	Rotten	9
10	Short	5	10	Dairy	7
11	Vegetative	5	11	Rice	6
12	Unfresh	4	12	Illness	6
13	Cardboard	4	13	Bitter	5
14	Weighty	4	14	Nutty	5
15	Unclean	3	15	Short	5
16	Rice	3	16	Cardboard	5
17	Popcorn	3	17	Bready	5
18	Dairy	3	18	Unfresh	4
19	Bready	3	19	Unclean	4
20	Drying	3	20	Drying	4
21	Rotten	2	21	Vegetative	3
22	Cereal	2	22	Yeasty	2
23	Yeasty	2	23	Spicy	2
24	Spicy	2	24	Flat	2
25	Flat	2	25	Roasted	2
26	Oxidation	2	26	Cereal	1
27	Roasted	2	27	Meaty	1
28	Meaty	1	28	Umami	1
29	Illness	0	29	Weighty	0
30	Umami	0	30	Oxidation	0
	Total	148		Total	206

Appendix 6: Answer sheets for EyeQuestion

Questionnaire for all assessors prior to T1

Gender:

- ☐ Female
- ☐ Male

Age:

- ☐ 18–29
- ☐ 30–39
- ☐ 40–49
- ☐ 50–59
- ☐ 60+

Experience (choose the option that represents the highest level of education):

- ☐ Completed WSET 3/Sommelier exam
- ☐ Completed WSET Diploma
- ☐ Completed Master of Wine

How often does your current position involve tasting wine? (Choose one response option)

- ☐ Daily
- ☐ Several times per week
- ☐ Weekly/several times per month
- ☐ Less frequently

In what part of the wine business do you mainly work (Choose one response option)?

- ☐ Wine journalism
- ☐ Wine education
- ☐ Wine imports
- ☐ The Norwegian Wine Monopoly

Do you currently have a cold or other factors that could influence your ability to smell and taste?

- ☐ Yes
- ☐ No

Comment:

Answer sheet during T1 and T2

Instructions:

- Taste the samples from left to right
- One of the samples is mousy
- Indicate which sample is mousy
- If you cannot detect mousiness in any of the samples, please guess
- Taste each sample with a minimum two-minute break between each tasting

Please indicate the mousy sample:

845 (1)	768 (2)	976 (3)
------------	------------	------------

Comment:

Description test of two wines containing C4 and C8 respectively, completed after T2

You have two glasses of wine in front of you. Please taste and evaluate the two wines from left to right with a break of a minimum of two minutes in between. Please describe each wine in your own words.

443:

782:

Questionnaire during the training period for the training group

Name:

What date are you training (choose one)?:

- ☐ Saturday 13 Jan.
- ☐ Sunday 14 Jan.
- ☐ Monday 15 Jan.
- ☐ Tuesday 16 Jan.
- ☐ Wednesday 17 Jan.
- ☐ Thursday 18 Jan.
- ☐ Friday 19 Jan.
- ☐ Saturday 20 Jan.
- ☐ Sunday 21 Jan.
- ☐ Monday 22 Jan.
- ☐ Tuesday 23 Jan.
- ☐ Wednesday 24 Jan.
- ☐ Thursday 25 Jan.
- ☐ Friday 26 Jan.

Bottle 1 (C4): Can you detect mousiness?

Indicate on the following scale your perception of mousiness: No mousiness (1) – (9) Strong mousiness.

Bottle 2 (C8): Can you detect mousiness?

Indicate on the following scale your perception of mousiness: No mousiness (1) – (9) Strong mousiness.

Comments/observations:

Questionnaire completed by the assessors in the training group after completing T2 and the description test (number of responses of the assessors indicated in parentheses)

Please answer the following questions:

How has the 14-day training period affected your personal acceptance of mousiness in wine?

- I now have a higher level of acceptance of the taint (e.g. you think mousiness is less unpleasant now than before you started the training) (1)
- I now have a lower level of acceptance of the taint (e.g. you think mousiness is more unpleasant now than before you started the training) (16)
- It has not affected my level of acceptance (6)

What is your personal perception of how the training has affected your sensitivity to mousiness?

- The training has increased my sensitivity to mousiness in wine (17)
- The training has decreased my sensitivity to mousiness in wine (2)
- It has not affected my sensitivity to mousiness in wine (4)

How has the training period affected how certain you are about your ability to detect mousiness in wine?

- The training has made me more certain about how to detect mousiness in wine (20)
- The training has made me less certain about how to detect mousiness in wine (0)
- It has not affected how certain I am about my ability to detect mousiness in wine (3)

How has the training affected your awareness regarding mousiness in wine?

- I now have a higher awareness of mousiness in wine (19)
- I now have a lower awareness of mousiness in wine (0)
- It has not affected my awareness of mousiness in wine (4)

Please describe in your own words your experience of being part of this study regarding both the training period and the tastings: